

Effects of Subsidies on Welfare and Market Structure in the U.S. Broadband Industry*

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Abstract

How should governments split broadband subsidies between consumer discounts and infrastructure grants? I study the Affordable Connectivity Program (ACP) and the Broadband Equity, Access, and Deployment (BEAD) program using state-year event studies and a tract-level structural model with endogenous entry. ACP was followed by lower prices, faster plan offerings, and lower concentration in high-eligibility states. BEAD produced smaller, delayed effects, consistent with construction and entry lags. I then use FCC Form 477, Urban Rate Survey, and ACS data to estimate demand and entry for 220,814 products in 69,085 tracts. Counterfactuals imply a BEAD benefit-cost ratio of 3.5–12.7 and an ACP ratio of 1.66–4.04. ACP is most valuable where broadband exists but remains unaffordable; BEAD is most valuable where infrastructure is missing. The programs are complements, not substitutes.

JEL classification: D22, L13, L51, L52, L96, L98.

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1 Introduction

Broadband access has become a prerequisite for economic participation, yet the digital divide remains wide. In 2020, roughly 21 million American households lacked fixed broadband service, a gap that falls disproportionately on low-income and rural communities.¹ Congress responded with the Infrastructure Investment and Jobs Act (IIJA) of 2021, which allocated approximately \$65 billion to broadband and created two flagship programs: the Affordable Connectivity Program (ACP), a \$14.2 billion demand-side subsidy providing eligible low-income households with monthly discounts of up to \$30 on broadband bills, and the Broadband Equity, Access, and Deployment (BEAD) program, a \$42.5 billion supply-side grant funding deployment in unserved and underserved areas. The two programs share a goal—wider, more affordable broadband—but attack different constraints. ACP lowers the consumer-facing price where service already exists. BEAD lowers the fixed cost of building service where private deployment has not materialized.

In this paper, I ask three questions. First, did ACP and BEAD change broadband prices, product offerings, and local market structure after IIJA? Second, how large are the welfare gains from each program once firms can enter or adjust local product positions? Third, are demand-side and supply-side subsidies substitutes, or do they address different parts of the digital divide? These questions matter for current policy. ACP exhausted its funding in June 2024, leaving policymakers to decide whether a successor affordability program is worth financing. BEAD is the largest federal broadband deployment program in U.S. history, but its returns depend on whether subsidies induce entry in markets where deployment is privately unprofitable. Standard evaluations that count new subscribers or dollars per connection do not answer these questions, especially in local broadband markets with few providers and substantial infra-marginal households.

To answer them, I combine reduced-form and structural evidence. The reduced-form analysis uses continuous-treatment event studies to compare states with different exposure to ACP eligibility and BEAD funding before and after IIJA. This design is well suited for measuring the timing and direction of observed policy effects. The structural model then asks what those effects imply for welfare and counterfactual policy design. I estimate a nested-logit demand system, recover Bertrand-Nash markups and marginal costs, and estimate fixed costs of entry using the scalable discrete-game bounds of [Fan and Yang \[2024\]](#). This approach is appropriate here because broadband policy changes both demand and product availability. It also avoids an infeasible full solution of the entry game during estimation, which is essential in

¹[Federal Communications Commission \[2020\]](#) reported 21.3 million Americans lacking access to fixed broadband at 25/3Mbps speeds based on FCC Form 477 data.

a setting with tens of thousands of local markets and many potential provider-speed positions.

The data combine several official federal sources. For the reduced-form analysis, I build a balanced panel of 52 U.S. states from 2014 to 2025 using FCC Urban Rate Survey prices and product characteristics, ACP eligibility exposure from the American Community Survey, and BEAD allocations from federal program records. For the structural analysis, I construct a pre-IIJA local product-level dataset for December 2020. The sample contains 220,814 broadband products in 69,085 census tracts, defined at the tract-provider-speed-tier level. Product availability comes from FCC Form 477 deployment records, prices from the FCC Urban Rate Survey, adoption and outside-option shares from FCC broadband adoption data and the American Community Survey, and demographics from ACS tract files. The data are not perfect—product-level subscriber counts are inferred and prices are matched from advertised URS plans—but they are unusually rich for this market: they observe local availability, prices, technology, demographics, and potential entry positions before either major IIJA program was implemented.

The results show that the programs operate through different margins. In the event study, ACP is associated with lower prices, faster plan mixes, and lower concentration in high-eligibility states. Moving from the 25th to the 75th percentile of ACP eligibility exposure is associated with broadband prices that were roughly 3–5 percent lower for average plans and somewhat more for high-speed plans at peak response. BEAD effects are smaller and appear with a two-to-three year lag, consistent with the time required for infrastructure deployment. In the structural counterfactuals, a 75 percent BEAD fixed-cost subsidy induces 115,000–341,000 new provider-tract entries and raises consumer surplus from \$2.18–\$2.65 billion per month to \$2.86–\$3.36 billion. A \$30 ACP discount raises consumer surplus to \$5.31–\$5.91 billion per month by expanding take-up at lower consumer-facing prices. In cost-benefit terms, BEAD generates a BCR of 3.5–12.7, while ACP generates 1.66–4.04. The central lesson is that the programs are complements: ACP is most useful where infrastructure exists but affordability limits adoption; BEAD is most useful where infrastructure is missing.

The paper contributes to three related literatures. First, it contributes to work on broadband markets and the digital divide. Existing studies show that broadband adoption is shaped by prices, income, local market structure, and provider choice [Rosston et al., 2010, Nevo et al., 2016, Flamm and Varas, 2022, Espín and Rojas, 2024, Kearns, 2024]. I build on this evidence but study policy exposure as a source of variation in both affordability and market outcomes.² Second, the paper contributes to structural work on differentiated-product

²Concentration, high prices, and limited choice are well documented in U.S. broadband [Flamm and Varas, 2022]. Downstream consequences for labor markets, education, and health are documented in Forman et al. [2005], Goldfarb and Prince [2008], and Goolsbee and Klenow [2002].

demand and endogenous product entry. The approach is closest in spirit to [Tamer \[2003\]](#), [Ciliberto and Tamer \[2009\]](#), and [Fan and Yang \[2024\]](#), and brings scalable discrete-game methods to broadband markets. Third, the paper contributes to the evaluation of public subsidies in imperfectly competitive markets. I compare a recurring demand-side subsidy and a fixed-cost deployment subsidy in the same welfare accounting exercise, separating affordability from infrastructure deployment. This comparison also speaks to work on public provision and infrastructure crowd-out, including [Wilson \[2025\]](#), who finds that municipal broadband can reduce private investment. I do not model that crowd-out channel for BEAD, so the results should be read as the welfare return to subsidized private entry rather than to public provision.

The rest of the paper proceeds as follows. Section 2 describes the institutional setting, data, and local market structure. Section 3 presents the reduced-form evidence. Section 4 develops the structural model, including timing assumptions, identification, and a discussion of multiple equilibria. Section 5 reports estimation results and structural parameter diagnostics. Section 6 presents counterfactual policy simulations and distributional effects across market types. Section 7 reports the cost-benefit analysis, including sensitivity to discount rate and infrastructure horizon. Section 8 concludes.

2 Industry Background and Data

2.1 Industry Background

2.1.1 Current Landscape

The U.S. broadband industry has grown substantially over the past two decades and now touches nearly every dimension of household economic life. This paper focuses on residential broadband, which has attracted sustained federal attention and large public investments. Broadband is offered in several forms, with internet service providers (ISPs) competing through differences in speed, data limits, and pricing.

Traditionally, local telephone and cable companies have been the primary providers of broadband services. Telephone companies deliver broadband through Digital Subscriber Line (DSL) technology, utilizing older copper wires installed on telephone poles. Cable companies provide broadband over coaxial cables originally deployed for television. Today, the industry has expanded with more firms, including large nationwide providers and smaller local operators. Local governments participate in the market in most states, except in a few

states where legislation restricts municipal provision.³ Most U.S. households connect through one of the following technologies: DSL, cable, fixed wireless, or fiber optics.⁴ Fixed wireless transmits the internet via radio signals from a fixed tower to a receiver at the customer’s location, making it practical for rural areas where wired infrastructure is costly. Fiber-optic broadband uses thin glass or plastic strands to transmit data as light pulses, offering superior speed, reliability, and bandwidth. It supports symmetrical upload and download speeds, low latency, and high-demand applications such as streaming, cloud services, and remote work. Fiber deployment, however, requires substantial investment and is primarily available in urban and suburban areas.

The literature on U.S. broadband consistently returns two findings: service is expensive relative to household income, and in most local markets consumers face limited provider choice [Flamm and Varas, 2022]. In my December 2020 sample, the average census tract has 2.36 providers offering fixed broadband, and the median provider HHI—computed within tracts using provider market shares—is 5,712, well above the 2,500 threshold traditionally associated with concentrated markets. Fiber-optic coverage, which offers the best performance and reliability, is concentrated in urban and suburban areas; rural tracts disproportionately rely on DSL over aging copper infrastructure, which has lower speeds and reliability.

These market structure conditions underlie what the federal government calls the digital divide: the gap between households with reliable internet access and those without. Income, geography, and digital literacy all predict which side of that gap a household falls on, and the consequences extend to job search, healthcare, and schooling [Forman et al., 2005, Goldfarb and Prince, 2008, Goolsbee and Klenow, 2002]. The federal government has responded with a range of subsidy programs: some target affordability directly; others subsidize infrastructure deployment in areas where private investment has not materialized.

2.1.2 Federal Broadband Subsidies

Congress passed the Infrastructure Investment and Jobs Act (IIJA) in 2021, allocating roughly \$65 billion to broadband. The two main programs are the Affordable Connectivity Program (ACP), which subsidized household internet bills directly, and the Broadband Equity, Access, and Deployment (BEAD) program, which funds deployment in underserved areas. Of the total, \$14.2 billion went to ACP and \$42.45 billion to BEAD.

ACP was introduced as a successor to the Emergency Broadband Benefit (EBB). The

³For instance, Texas state law significantly restricts municipal governments from directly providing broadband services to residents (see Baller, Stokes and Lide, 2020; BroadbandNow, 2023). Wilson [2025] finds that municipal entry crowds out private ISP investment, suggesting that public provision involves a market-structure trade-off beyond the deployment-gap rationale that motivates programs like BEAD.

⁴See Appendix A for illustrative images.

EBB program, which ran from February to late 2021, reached nearly nine million low-income households with a subsidy of up to \$50 per month. The ACP enrolled 23 million households at its peak, offering a monthly discount of up to \$30 (\$75 on qualifying Tribal lands). Eligible households could also receive a one-time discount of up to \$100 to purchase a connected device.⁵ ACP funding was exhausted in June 2024 when appropriations were not renewed by Congress. The program’s termination is particularly relevant for this paper: the period 2022–2024 represents ACP’s full operational lifecycle, from launch through exhaustion, and is fully covered by the reduced-form event study panel. ACP’s termination also raises the immediate policy question of whether a successor program is warranted and, if so, what form it should take. The welfare and fiscal-efficiency analysis in this paper directly informs that debate.

The BEAD program represents the largest federal investment targeting broadband providers directly. It allocates funds to states based on the number of underserved locations in each jurisdiction. The FCC defines underserved locations as broadband-serviceable locations that either lack broadband service entirely or lack access to reliable service at speeds of at least 25/3 Mbps with latency below 100 milliseconds.⁶ Allocations are determined by federal formulas, so states cannot independently influence the amount of funding they receive. Although IIJA was enacted in November 2021, state-level BEAD allocations were not announced until June 2023, and states were required to submit initial proposals by December 2023. This two-year gap between enactment and allocation announcement—followed by subgrantee selection, permitting, and construction—directly explains the two-to-three year lag in market effects documented in the event study. As of 2025, states were in various stages of implementation, with deployment expected to extend through 2026 and beyond. The structural model uses a pre-IIJA baseline and provides counterfactual predictions that are best understood as the program’s steady-state equilibrium effects once deployment is complete, rather than as predictions of specific intermediate-year outcomes.

2.1.3 Economic Rationale

The economic rationale for both programs rests on market failures in residential broadband. On the demand side, low income and limited digital literacy create participation barriers: even where broadband is available, many households cannot afford monthly plans. The ACS reports that in 2020, roughly 15 percent of U.S. households with broadband access reported being unable to afford their current plan, and broadband subscription rates among households below 200 percent of the federal poverty line were roughly 20 percentage points below the

⁵<https://www.fcc.gov/acp>

⁶<https://broadbandusa.ntia.doc.gov>

national average. A demand-side subsidy addresses this access gap directly, lowering the effective price to a level where price-constrained households are willing to subscribe.

On the supply side, the market failure is different: fixed deployment costs are high enough to make broadband deployment privately unprofitable in low-density or low-income areas, even when the social benefits of connectivity are large. The tension between private and social returns to infrastructure investment is a classic rationale for public subsidy [Bresnahan and Reiss, 1991, Kearns, 2024]. An important caveat is that public supply-side intervention may also crowd out private entry: Wilson [2025] finds that municipal broadband provision reduces private ISP investment in affected markets, a general-equilibrium cost that standard program evaluations omit. BEAD directly addresses the deployment margin by reducing the provider’s private cost of entry, but the net welfare effect depends on how much private investment would have occurred in its absence.

The economic distinction matters for program design and evaluation. A demand-side subsidy like ACP has no direct effect on whether infrastructure is available in the first place; it can only stimulate adoption in markets where ISPs already offer service. In unserved areas—where infrastructure does not exist—no level of demand-side subsidy can induce adoption. BEAD, by contrast, targets the deployment decision directly. But BEAD cannot improve affordability in markets where infrastructure already exists and prices are the binding constraint. The two programs are thus economic complements: each addresses a market failure the other cannot.

An additional consideration is that the broadband market is oligopolistic, not perfectly competitive. Most U.S. households have access to at most two or three fixed-broadband providers, and many face a single dominant ISP [Flamm and Varas, 2022]. In this setting, demand-side subsidies may affect more than household take-up: a subsidy that expands the addressable market can also be associated with changes in posted prices and product offerings. This possibility motivates the reduced-form analysis, while the structural counterfactuals below keep ACP’s direct demand-expansion margin analytically separate from any additional provider price adjustment.

2.2 Data Sources and Estimation Samples

The empirical analysis uses two complementary samples. The first is a state-year policy panel used to estimate reduced-form effects of federal broadband funding exposure. The second is a newly constructed tract–provider–product–level sample used to estimate demand, recover marginal costs, and discipline the structural counterfactuals. The state-year panel combines IIJA and BEAD allocation records with annual broadband prices, product characteristics, and

state demographics. The tract level sample combines FCC Urban Rate Survey prices, FCC Form 477 deployment records, FCC broadband adoption data, and American Community Survey demographics.

The state-year specification is motivated by both identification and data quality. Constructing a consistent tract-year panel over 2014–2025 is complicated by the FCC’s transition from Form 477 to the Broadband Data Collection (BDC) system after 2021, which substantially altered geographic reporting units and coverage measurement. The state-year design exploits policy exposure variation while avoiding spurious dynamics induced by this methodological break. The structural model, which uses a single cross-section predating IJJA, is unaffected by this transition.

For the demand and supply estimation, a market is a census tract and a product is a canonical provider⁷ $\times$ speed-tier pair. The baseline sample is constructed for December 2020, before the implementation of the major IJJA broadband programs. Speed tiers follow the IJJA broadband standard: the high-speed tier contains products with download speed of at least 25 Mbps and upload speed of at least 3 Mbps; the low-speed tier contains all other fixed broadband products. Appendix B reports file-level details, merge keys, sample restrictions, and data-source links. The full replication package is available upon request.

2.3 Urban Rate Survey Price Matching

Monthly prices are assigned to each observed and potential broadband product using plan-level observations from the FCC Urban Rate Survey (URS). Similar price assignment strategies using the URS have been adopted in [Espín and Rojas \[2024\]](#) and [Kearns \[2024\]](#).⁸ Products are defined at the census-tract-canonical-provider-speed-tier level. URS prices are converted to constant 2025 dollars using CPI adjustment factors.

For each product, the assignment procedure searches for comparable URS plans using a six-level brand-first hierarchy. The algorithm first searches for plans in the same state, offered by the same canonical provider, and in the same speed tier. If no such plan is available, the search expands sequentially to national provider-tier matches, provider-specific matches that ignore the speed tier, state-level tier matches pooling across providers, and finally national tier-level matches across all providers.

⁷Raw FCC provider names contain subsidiary, regional, and legacy operating names belonging to the same corporate family. I consolidate these into a normalized corporate-family identifier before constructing products, assigning prices, calculating market shares, and analyzing entry. For example, Southwestern Bell, Indiana Bell, and BellSouth are mapped to `att`; Time Warner Cable, Bright House, and Spectrum to `charter`; CenturyLink, Quantum Fiber, and Lumen to `centurylink`; and Starlink and SpaceX to `starlink`.

⁸Both papers match FCC Form 477 deployment records to URS plans by provider, state, and speed tier, assigning the advertised monthly price of the closest-speed plan as the product price—the same logic as the hierarchy used here.

Within the first non-empty match category, the assigned price is the URS **Total Charge** of the plan whose advertised download speed is closest to the FCC product’s average advertised download speed. The procedure additionally records the matched URS plan speed, the percentage distance between FCC and URS plan speeds, the match level used, and an indicator for successful price imputation.

The same assignment rule is applied to both observed and potential products. For observed products, average advertised download speed is computed from Form 477 block-level deployment records aggregated to the tract–provider–speed–tier level. For potential products, characteristics are assigned using the provider’s state-level average within the corresponding speed tier. Products without a valid URS match are retained with missing prices and flagged as unmatched.

The hierarchy achieves a high degree of specificity in practice. For the December 2020 sample, 40.8% of products are matched at the most granular level (state $\times$ brand $\times$ tier), while 75.5% are matched using state-specific information. Only 1.4% of products rely on the broadest national-tier fallback. Overall, the procedure achieves a 100% price imputation rate for the 673,110 product observations in the estimation sample. Additional implementation details are provided in Appendix D.

2.4 Market Definition and Sample Construction

A market is a U.S. census tract. This definition is both fine enough to capture genuine geographic heterogeneity in competition and coarse enough to be measurable with publicly available data. Census tracts contain on average 4,000 residents and are designed by the Census Bureau to have roughly homogeneous socioeconomic characteristics. Broadband markets have natural local boundaries: ISPs compete for households within a given geography, and the cost of expanding infrastructure is increasing in distance. Tract-level markets have become the standard unit of analysis in empirical work on local broadband competition [Flamm and Varas, 2022, Kearns, 2024].

A product is a canonical-provider–speed–tier pair within a tract. I consolidate provider names to canonical corporate families (e.g., all AT&T subsidiaries to AT&T) and define speed tiers using the IIJA broadband standard: the high-speed tier ($g = H$) contains products with download speed of at least 25 Mbps and upload speed of at least 3 Mbps; the low-speed tier ($g = L$) contains all other fixed broadband offerings. This two-tier classification captures the most policy-relevant distinction—products meeting versus not meeting the federal minimum broadband standard—while remaining tractable for the entry game.

Market size M_t is the number of households in the tract from the 2020 American Commu-

nity Survey. The inside market (broadband subscribers) and outside option (non-subscribers) are defined relative to this population.

2.5 Constructing Product-Level Market Shares

The central measurement challenge in estimating broadband demand is the absence of publicly available product-level market shares. The FCC reports where providers offer service, the technologies they use, and advertised speeds, but does not publicly report subscribers by provider-speed product in each census tract. This missing object is essential for demand estimation.

I construct product-level shares by combining FCC adoption information with tract product availability and characteristics, following the logic of Crawford et al. [2019]. The FCC reports tract-level fixed broadband adoption by speed threshold in binned ranges. For example, a tract in the $400 < x \leq 600$ bin has an adoption rate between 40 and 60 percent for the relevant fixed broadband speed category.⁹

I use these binned adoption measures to recover aggregate adoption by speed tier. Let S_t^H denote the high-speed adoption share in tract t , and S_t^L the low-speed adoption share. In the baseline construction, I assign S_t^H using the midpoint of the FCC adoption interval, and recover low-speed adoption as the residual:

$$S_t^L = S_t^{\text{fixed}} - S_t^H, \quad (2.1)$$

where S_t^{fixed} is total fixed-broadband adoption share in tract t available in the American Community Survey. The outside-option share is

$$s_{0t} = 1 - S_t^H - S_t^L. \quad (2.2)$$

The next step allocates tier-level adoption across providers available in the tract-provider-tier level. I first estimate an aggregate adoption equation at the tract-tier level:

$$\log\left(\frac{S_t^{\text{fixed}}}{s_{0t}}\right) = \bar{X}_t' \kappa + D_t' \lambda + u_t, \quad (2.3)$$

⁹See, for example, FCC, *Residential Fixed High-Speed Connections per 1,000 Households by Census Tract, FCC Form 477 Data as of December 31, 2008*, <https://docs.fcc.gov/public/attachments/DOC-296239A2.pdf>. For the 2020 broadband environment, see FCC, *Internet Access Services: Status as of December 31, 2020*, <https://docs.fcc.gov/public/attachments/DOC-395959A1.pdf>. The FCC also documents that fixed broadband subscription data are reported by census tract, technology, and advertised speed; see <https://help.bdc.fcc.gov/hc/en-us/articles/5296902403867-How-to-Format-Fixed-Broadband-Subscription-Data>.

where $\bar{X}_t$ contains tract-level averages of product characteristics and D_t contains ACS demographic controls in . Within each tract and speed tier, I allocate the FCC-implied adoption share across available products using logit shares implied by predicted utilities:

$$\hat{s}_{j|g,t} = \frac{\exp(\hat{\delta}_{jgt})}{\sum_{\ell \in \mathcal{J}_{gt}} \exp(\hat{\delta}_{\ell gt})}, \quad (2.4)$$

where $\mathcal{J}_{gt}$ is the set of products in tier g available in tract t . The product-level market share is then

$$\hat{s}_{jgt} = \hat{s}_{j|g,t} S_t^g. \quad (2.5)$$

By construction, product shares sum to total fixed-broadband adoption in each tract. The constructed shares should be interpreted as inferred product shares, not directly observed subscriber counts. Appendix E describes the treatment of FCC adoption bins, the variables used in (2.3), and diagnostics for the share-construction procedure.

Table 1 reports summary statistics for the demand and supply sample used in estimation.

Table 1: Summary Statistics: Demand and Supply Sample

Variable	Mean	Std. Dev.	P25	P75
<i>Product-level variables</i>				
Price (\$/month)	71.65	25.34	49.99	86.17
Download speed (Gbps)	0.337	0.713	0.025	0.300
Upload speed (Gbps)	0.065	0.136	0.003	0.030
High-speed indicator	0.624	—	—	—
Product market share $\hat{s}_{jt}$	0.259	0.152	0.159	0.324
Within-tier share $\hat{s}_{j g,t}$	0.313	0.166	0.202	0.383
<i>Tract-level variables</i>				
Outside-option share s_{0t}	0.171	0.134	0.100	0.300
Products per tract	3.20	—	2	4
Providers per tract	2.36	—	2	3
Provider HHI	5,712	2,584	3,694	6,394
Households	1,679	820	1,126	2,089
Population	4,511	2,325	2,967	5,601
Median household income (\$)	67,152	33,805	44,318	82,159
Renter share	0.364	0.228	0.183	0.509
College share	0.311	0.193	0.161	0.425
Age 65+ share	0.164	0.080	0.114	0.200
Broadband adoption share	0.668	0.171	0.559	0.803

Notes: The sample contains 220,814 tract-provider-tier products in 69,085 census tracts and 1,732 canonical providers. The high-speed tier is defined as download speed of at least 25 Mbps and upload speed of at least 3 Mbps. Product shares are inferred using FCC adoption information and product-level availability (Section 2.5). Provider HHI is computed within tracts using provider shares among fixed-broadband subscribers and is scaled from 0 to 10,000.

2.6 Broadband Market Structure

The market structure of the broadband industry shapes both the identification strategy and the interpretation of the structural results. Table 2 summarizes the distribution of provider counts in the estimation sample.

Nearly a quarter of census tracts in the sample have a single fixed-broadband provider, and almost two-thirds have at most two. This concentration is one of the market failures the federal programs address. Monopoly tracts account for 21.96 percent of households—a substantial fraction facing no intra-market price competition for fixed broadband. The moderate share of tracts with three or more providers (36.73 percent) reflects that competitive broadband markets, while they exist, are not the norm.

Table 2: Provider Competition in the Sample

Number of providers in tract	Share of tracts (%)	Share of households (%)
One provider	23.10	21.96
Two providers	40.16	39.22
Three or more providers	36.73	38.82

Notes: Provider counts based on canonical providers with positive constructed product shares in the tract. Sample: 69,085 census tracts in December 2020.

Table 3 documents the structure of national demand. The top nine providers jointly account for 74.3 percent of implied subscribers in the sample, with AT&T and Comcast alone holding roughly 46 percent. This concentration at the national level contrasts with fragmented local markets: while a small number of large providers dominate nationally, their geographic footprints do not overlap sufficiently to create competitive local markets. This geographic segmentation is precisely why supply-side deployment subsidies like BEAD—targeted at specific unserved locations rather than national incumbents—are better suited to the entry margin than demand-side programs.

Table 3: National Market Shares by Provider

Provider	Implied subscribers (Millions)	Share of inside good (%)	Share of households (%)	Tracts
AT&T	27.49	28.21	23.69	33,344
Comcast	17.28	17.74	14.90	28,950
Verizon	10.53	10.80	9.07	12,070
Frontier	7.83	8.03	6.75	9,549
Cox	3.77	3.87	3.25	5,540
Windstream	1.99	2.04	1.71	2,946
Consolidated	1.26	1.29	1.08	2,035
Starry	1.16	1.19	1.00	2,671
Optimum	1.13	1.16	0.98	2,670
All others (<1% each)	25.02	25.67	21.56	—

Notes: Provider shares computed by aggregating implied subscribers across census tracts and speed tiers. “Share of inside good” is the provider’s share among fixed-broadband subscribers. “Share of households” is relative to total Census households in the sample. All figures reflect the December 2020 baseline, which predates major IJJA broadband programs. Starry Internet went out of bankruptcy in February 2023 and subsequently exited the market; its presence here reflects its operational status as of the sample date.

3 Reduced-Form Evidence

3.1 State-Year Panel and Treatment Variables

The reduced-form analysis uses a balanced panel of 52 U.S. states (including Puerto Rico and the District of Columbia) observed annually from 2014 to 2025, yielding $N = 624$ state-year observations before missing values. The outcomes are drawn from the FCC Urban Rate Survey (URS), which records plan-level prices, speeds, and technology types at the state level each year. I aggregate to a state $\times$ year panel and compute weighted-average prices (weights equal the number of plans) and technology shares. Real prices are deflated to 2025 USD using the CPI-U. I examine thirteen outcomes organized into three groups: prices (log real prices by speed tier), technology and speed (upload speed, fiber/DSL/fixed-wireless shares), and market structure (provider counts, entry, exit, HHI, and top-provider share). Table 4 reports descriptive statistics.

Table 4: Descriptive Statistics

Outcome	N	Mean	SD	Min	Median	Max
<i>Prices</i>						
Log avg. real price	597	4.583	0.190	3.827	4.556	5.254
Log price (<25/3 Mbps)	563	4.200	0.210	3.314	4.223	5.073
Log price ($\geq$ 25/3 Mbps)	587	4.703	0.223	4.169	4.673	5.483
<i>Technology & Speed</i>						
Upload speed (Gbps)	597	0.231	0.355	0.001	0.094	2.667
Fiber share	597	0.206	0.205	0.000	0.168	1.000
DSL share	597	0.206	0.250	0.000	0.108	1.000
Fixed-wireless share	597	0.039	0.137	0.000	0.002	1.000
<i>Market Structure</i>						
Provider count	597	6.754	5.762	1.000	5.000	49.000
Provider $\times$ technology count	597	8.918	8.050	1.000	7.000	56.000
Provider entries	597	3.451	3.408	0.000	3.000	32.000
Provider exits (next year)	597	3.451	3.881	0.000	2.000	30.000
Provider HHI (supply)	597	0.524	0.238	0.146	0.477	1.000
Top-provider share	597	0.637	0.209	0.225	0.613	1.000

Notes: Unit of observation is state $\times$ year, 2014–2025 ($N = 624$ state-year cells before missing). Prices are weighted-average plan prices deflated to 2025 USD using the CPI. Shares and the HHI are bounded $[0, 1]$.

Panel structure and outcome construction. The state-year panel spans 2014–2025, providing five pre-ACP years (2014–2021), three ACP years (2022–2024), and one follow-up year (2025). For BEAD, the panel provides six pre-announcement years (2014–2020), one year

following IIJA enactment (2021), and four post-enactment years (2022–2025). This timing is useful for the event study design: five pre-period observations provide enough pre-trend tests to have power against violations of parallel trends, while four to five post-period observations are sufficient to identify differential dynamic effects.

The URS collects plan-level data from a stratified sample of urban census tracts annually. Each record reports a plan’s advertised download and upload speeds, monthly charges, and technology type. I compute state-year weighted averages and technology shares from these plan-level records. Because URS sampling targets urban areas, the state-year panel is best interpreted as capturing urban and suburban broadband market conditions. Rural broadband dynamics, which are the primary target of BEAD, are underrepresented in the URS-based outcomes. This means the event study estimates of BEAD effects are likely conservative: if BEAD’s price and concentration effects are disproportionately concentrated in rural areas (as the structural analysis suggests), URS-based outcomes will understate the aggregate BEAD impact. In contrast, ACP’s effects, which operate through demand expansion in served (predominantly urban and suburban) markets, are more directly captured by the URS sampling frame.

Treatment intensity variables are both time-invariant and predetermined. This design avoids the endogenous variation concerns that arise when treatment intensity is measured contemporaneously with outcomes. The ACP eligibility share ($FPL200_i$) is drawn from the 2020 ACS, before ACP was implemented. The BEAD intensity variable ($\log BEAD_i$) is determined by federal allocation formulas based on FCC-certified unserved location counts, which are also measured before BEAD grants were disbursed. Neither measure can be influenced by ISP behavior or state broadband outcomes over the event window, supporting the identifying assumption that differential treatment assignment is uncorrelated with differential trends.

ACP exposure. I measure state-level ACP eligibility exposure as the share of households with incomes below 200% of the federal poverty level ($FPL200_i$), drawn from the 2020 American Community Survey. This is a time-invariant, pre-determined cross-sectional measure that varies across states and proxies for the potential demand stimulus ACP can generate. The program became active in 2022; I set $Post_t^{ACP} = \mathbf{1}[t \geq 2022]$.

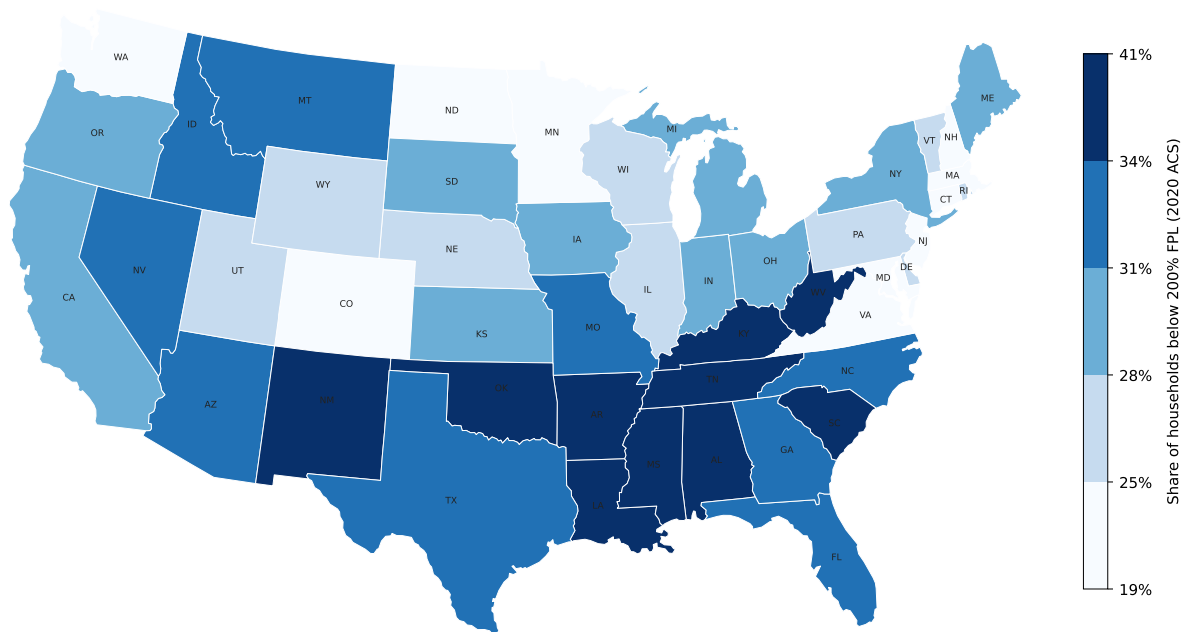


Figure 1: State-level ACP exposure: share of households with incomes below 200% of the federal poverty level ($FPL200_i$), measured from the 2020 ACS. Darker shading indicates higher ACP eligibility exposure. Colour breaks at sample quintiles.

BEAD exposure. I measure state-level BEAD intensity as the log of BEAD allocation (USD) per unserved broadband location ($\log BEAD_i$). A higher value indicates that a state receives more funding per targeted location, implying a more intensive supply-side intervention. This measure is also time-invariant and pre-determined by federal allocation formulas; states cannot influence their allocation. BEAD was announced in 2023; I set $Post_t^{BEAD} = \mathbf{1}[t \geq 2023]$.

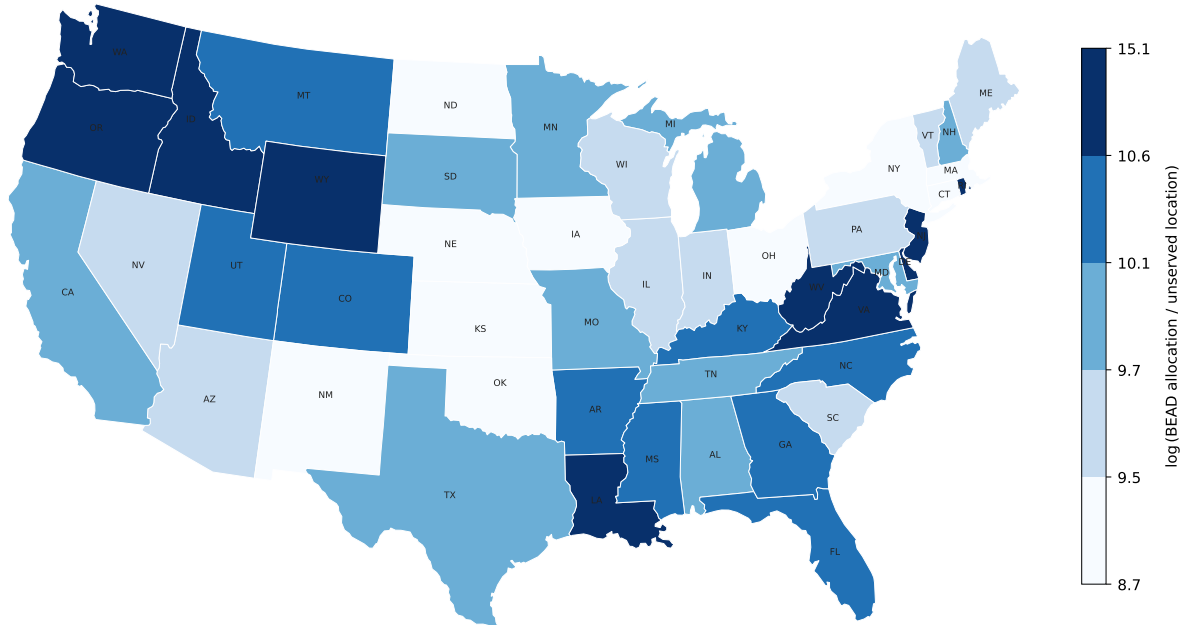


Figure 2: State-level BEAD exposure: log of BEAD allocation (USD) per unserved broadband location ($\log \text{BEAD}_i$), based on NTIA allocation announcements (June 2023). Darker shading indicates higher funding intensity per targeted location. Colour breaks at sample quintiles.

3.2 Identification Strategy

For program $P \in \{\text{ACP}, \text{BEAD}\}$, let D_i^P denote the time-invariant treatment intensity for state i and let τ index event time relative to the policy year. The dynamic specification is:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{\substack{\tau = -5 \\ \tau \neq -1}}^{\bar{\tau}^P} \beta_\tau^P \left(D_i^P \times \mathbf{1}[t = \bar{t}^P + \tau] \right) + \gamma_i' t + \varepsilon_{it}, \quad (3.1)$$

where α_i are state fixed effects, λ_t are year fixed effects, $\bar{t}^{\text{ACP}} = 2022$ and $\bar{t}^{\text{BEAD}} = 2021$. The omitted period is $\tau = -1$ (the year before the policy), so all estimates are relative to that baseline. Regressions are weighted by the number of plans n_{it} and standard errors are clustered at the state level.

The role of fixed effects. State fixed effects α_i absorb all time-invariant determinants of broadband outcomes at the state level, including permanent differences in geographic coverage, regulatory environment, and demographic composition. Identification therefore exploits *within-state* variation in outcomes over time, purged of any level differences across states. Year fixed effects λ_t absorb common national shocks to broadband markets, such as

economy-wide shifts in input costs, federal regulatory changes, or aggregate demand trends. The interaction of treatment intensity with year indicators then captures whether states more intensively exposed to each program moved differentially relative to the national trend.

State-specific trends. For both ACP and BEAD, I include state-specific linear time trends $\gamma_i' t$ to account for the possibility that treatment intensity correlates with pre-existing state-level broadband trajectories. For ACP, states with higher poverty rates—which mechanically have higher eligibility exposure—may have been following divergent broadband price and adoption paths before the program launched. For BEAD, states with more unserved locations tend to be lower-density and lower-income, and may have been experiencing systematically different infrastructure investment trends prior to IIJA enactment. The digital divide literature has documented that broadband adoption and pricing trends differ systematically across income and demographic groups [Goldfarb and Prince, 2008, Espín and Rojas, 2024]. Without state-specific trends, such pre-existing divergence would contaminate both sets of estimates. State-specific trends absorb smooth state-level deviations from the national year effects; the treatment coefficients $\hat{\beta}_\tau^{\text{ACP}}$ and $\hat{\beta}_\tau^{\text{BEAD}}$ are then identified from non-linear departures from each state’s pre-existing trend coinciding with program rollout.

Assumption 1 (Conditional parallel trends). *Conditional on state fixed effects, year fixed effects, and state-specific linear time trends, the untreated potential outcome follows the same trend across states with different treatment intensities D_i^P . Formally, $\mathbb{E}[\varepsilon_{it} \mid D_i^P, i, t] = 0$ for all $\tau < 0$.*

The identifying assumption is that states with higher treatment intensity D_i^P were not on systematically different trajectories for reasons unrelated to the policy. Because both FPL200_i and $\log \text{BEAD}_i$ are predetermined by pre-program demographic composition and federal allocation formulas, the main threat to identification would be that states with higher exposure were on different broadband trends before the programs launched. The pre-period coefficients $\hat{\beta}_\tau^P$ for $\tau < 0$ serve as direct falsification tests of Assumption 1.

A secondary concern is the anticipation of program benefits: ISPs or households might alter behavior before formal program launch in expectation of the policy. Under the IIJA timeline, BEAD was announced in November 2021 but allocations were not finalized until 2023, so anticipatory responses would appear as effects at $\tau = 1$ – 2 (2022–2023), which is largely consistent with what I find (early effects are near zero). For ACP, EBB, the predecessor program, ran from February to late 2021, meaning some anticipatory ISP behavior may already be absorbed in the $\tau = -1$ base period; this would lead me to understate ACP’s total effect. Neither concern appears to drive the main qualitative results.

3.3 ACP: Event Study Results

Figure 3 presents event study estimates from (3.1) for six key outcomes. Event time is indexed relative to 2022 ($\tau = 0$); the dashed vertical line marks November 2021 (IIJA enactment). The omitted period is $\tau = -1$ (year 2021).

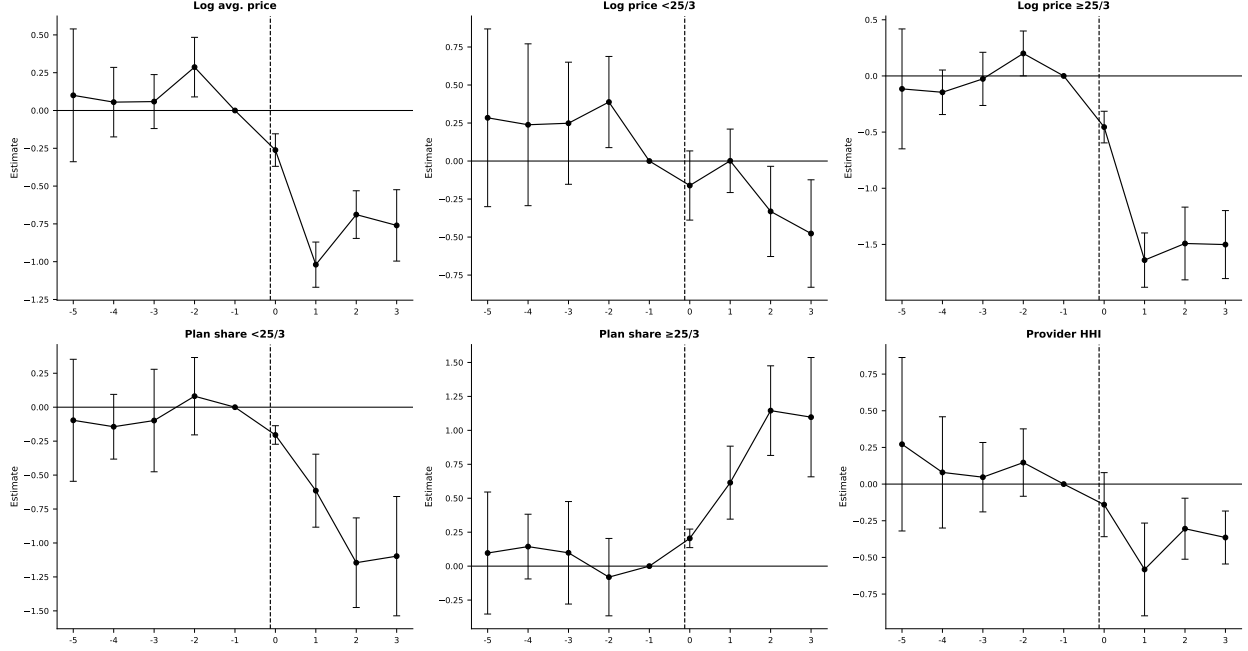


Figure 3: ACP event study estimates. Treatment intensity: share of households below 200% FPL. TWFE with state-specific linear trends, weighted by number of plans, standard errors clustered by state. Bars indicate 95% confidence intervals.

Pre-trends. The pre-period pattern is broadly consistent with the parallel-trends assumption, conditional on state and year fixed effects and state-specific linear trends. For most outcomes (log prices for high-speed plans, plan shares, and provider HHI), the pre-period coefficients are small and close to zero. Log price for plans below 25/3 Mbps shows somewhat elevated pre-period point estimates at $\tau = -5$ to $\tau = -3$, though with wide confidence intervals. This motivates the inclusion of state-specific linear trends, which absorb differential pre-existing trajectories in broadband prices across states at different poverty rates.

Post-ACP effects. A clear pattern emerges in the post-period. Average broadband prices fall significantly in higher-eligibility states following ACP's 2022 launch. The coefficient $\hat{\beta}_1^{\text{ACP}}$ for log average price is approximately -1.0 per unit of FPL200 share. To translate this into economically interpretable magnitudes, the interquartile range of state FPL200 shares in my panel is approximately 0.10 (from about 0.27 at the 25th percentile to 0.37

at the 75th). Moving from the 25th to the 75th percentile of ACP eligibility exposure is therefore associated with an additional log average price decline of roughly $-0.10 \times 1.0 \approx 10$ log points (approximately 10 percent) at peak ($\tau = 1$), stabilizing at roughly -7 percent through $\tau = 2-3$.

The price decline is especially pronounced for high-speed plans (log price $\geq 25/3$ Mbps), where the estimated response is approximately -0.15 to -0.18 log points per percentage-point increase in FPL200 share at $\tau = 1$, implying a 75th-to-25th percentile comparison of roughly 15–18 percent. This effect persists through $\tau = 2-3$. Plan composition shifts substantially: the share of low-speed plans (below 25/3 Mbps) contracts through $\tau = 2-3$, while the high-speed share expands commensurately, consistent with ISPs upgrading their product portfolios in response to elevated demand for faster service from newly subsidized households. Provider HHI declines in higher-exposure states, though the estimates are noisy and should be interpreted with caution.

The reduced-form evidence therefore points to more than a mechanical increase in subsidized subscriptions. In high-eligibility states, observed prices fell and the mix of plans shifted toward higher-speed offerings. I interpret this as evidence that ACP expanded the relevant customer base for ISPs, while leaving the exact decomposition of the price movement—pass-through, strategic pricing, or changes in plan composition—to the structural and institutional discussion below.

3.4 BEAD: Event Study Results

Figure 4 presents event study estimates for BEAD. Event time is indexed relative to 2021 (IIJA enactment, $\tau = 0$); the omitted period is $\tau = -1$ (year 2020). The specification follows (3.1): state and year fixed effects and state-specific linear time trends $\gamma'_i t$, weighted by number of plans, with standard errors clustered at the state level. State-specific trends are warranted for BEAD because states with more unserved locations tend to be lower-density and lower-income and may have been following systematically different infrastructure investment trajectories prior to IIJA enactment; the trends absorb these smooth pre-existing differentials so that the BEAD coefficients are identified from non-linear departures coinciding with program rollout.

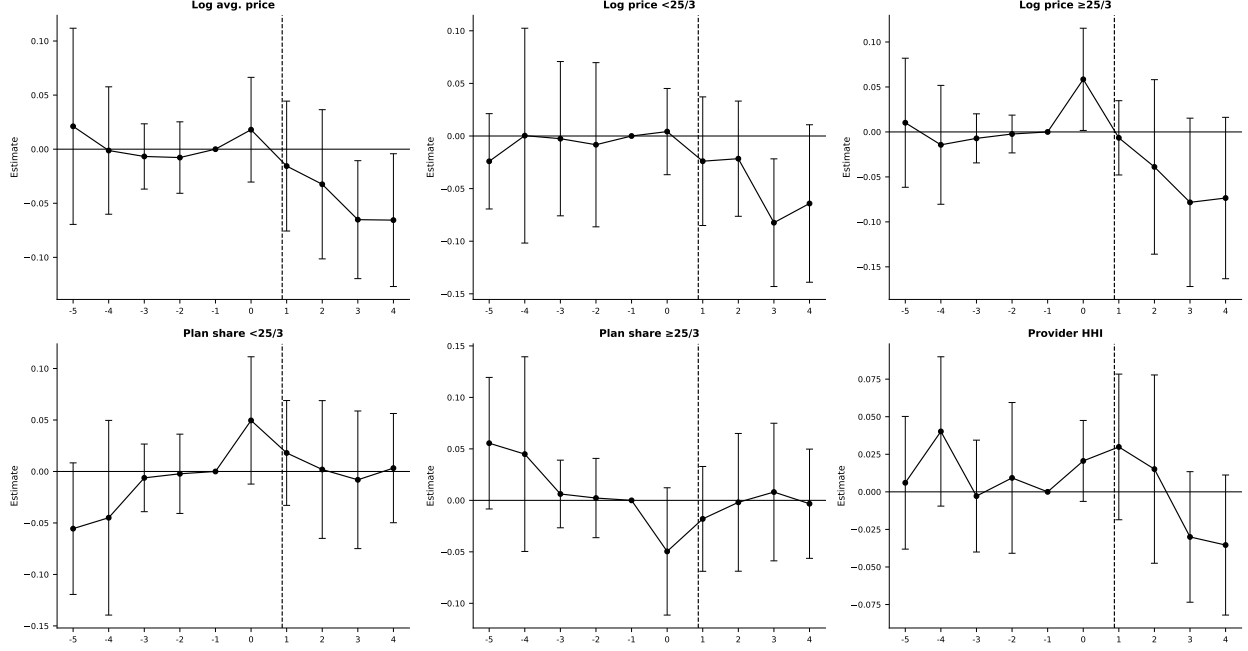


Figure 4: BEAD event study estimates. Treatment intensity: log BEAD allocation per unserved location. TWFE with state-specific linear trends, weighted by number of plans, standard errors clustered by state. Event time $\tau = 0$ corresponds to 2021 (IIJA enactment); the dashed vertical line marks $\tau = 1$ (2022), the first year following enactment. Bars indicate 95% confidence intervals.

Pre-trends. Pre-period coefficients are small and statistically indistinguishable from zero across all outcomes. States that ultimately received more BEAD funding per unserved location were on similar trajectories in prices, technology, and market structure before the program was announced.

Post-BEAD effects. BEAD produces smaller effects than ACP, and they materialize with a longer lag. In the first two years following IIJA enactment ($\tau = 1-2$, 2022–2023), effects on all outcomes are minimal and statistically indistinguishable from zero. Price responses become more apparent at $\tau = 3-4$ (2024–2025), with log average price declining by roughly 0.05–0.07 log points per log unit of BEAD intensity. Provider HHI shows a similarly lagged response, turning negative at $\tau = 3-4$ after being near zero or slightly positive in the first two post-enactment years. Plan composition, by contrast, shows no discernible shift over the event window: the shares of high-speed and low-speed plans are statistically indistinguishable from zero throughout $\tau = 1-4$. This timing fits the institutional rollout of BEAD. The event window captures only the early phase of infrastructure deployment: new networks must be built, interconnected, and activated before entry can discipline incumbent pricing.

3.5 Summary DiD Estimates

To complement the event study evidence, Table F.1 (Appendix F) reports static two-way fixed-effects difference-in-differences estimates for each outcome, pooling the post-period. These estimates summarize the average treatment effect over the post-program period and are directly comparable across outcomes and programs. They serve as a robustness check on the event study patterns: if the dynamic patterns are genuine, the static DiD should be significant and in the same direction as the post-period event study coefficients.

The DiD estimates tell the same story. For ACP, prices fall significantly in higher-eligibility states (log average price coefficient -0.503 , log high-speed price -0.990), fiber share rises ($+0.347$), DSL share falls (-0.442), and provider HHI declines (-0.148). These estimates are consistent with price and product-mix adjustments in higher-exposure states. For BEAD, estimates are smaller and less precisely estimated, consistent with the event study showing effects concentrated in the later post-enactment years. Provider HHI and top-provider share show modest negative effects, consistent with the lagged entry response documented in the event study. The absence of significant plan-composition effects for BEAD confirms that BEAD operates primarily through the entry margin rather than product repositioning.

3.6 What the Event Studies Identify and What They Cannot

The event studies leave two empirical facts. ACP generated significant, relatively immediate market responses: prices fell, product mix improved, and concentration declined. These facts suggest that demand-side subsidies can be associated with market adjustments beyond the direct household discount. BEAD produced smaller, lagged effects consistent with the capital-intensive nature of infrastructure deployment.

The event study is useful for the *direction, timing, and reduced-form magnitude* of policy effects, but three limitations matter for the rest of the paper.

First, the event study cannot separate ACP’s demand-expansion effect from other price-adjustment margins. The observed price declines are consistent with several channels, including direct pass-through, strategic price adjustment, and changes in the composition of available plans. The structural model quantifies demand expansion while holding gross provider prices fixed. The two exercises therefore answer different questions: the event study documents reduced-form movements in prices and product mix, while the structural model values the demand and entry channels it explicitly models.

Second, the event study is identified at the state level, while the structural model is estimated at the census-tract level. These differences in spatial resolution mean the two analyses are measuring different margins. State-level event studies capture average effects

across heterogeneous local markets; tract-level structural estimates allow for rich heterogeneity in market structure, provider presence, and cost conditions that the state panel cannot capture.

Third, the event study identifies only the effects of the programs as implemented. It cannot evaluate alternative policy designs—a 50 versus 75 percent BEAD subsidy rate, a \$10 versus \$30 ACP discount, or scenarios in which both programs operate simultaneously. Nor can it decompose welfare changes into consumer surplus, producer surplus, and fiscal cost, or identify the entry and product-choice responses that drive the observed price and concentration patterns. Those questions require the structural model developed in Sections 4–5.

The two analyses are therefore complements. The event study provides reduced-form evidence that the programs changed market outcomes. The structural model explains the channels it is built to capture, measures welfare, and enables policy counterfactuals.

Implications for policy evaluation. Standard program evaluations of demand-side subsidies count the number of newly connected households and calculate the cost per subscriber [Goolsbee and Guryan, 2006].¹⁰ That accounting is useful, but it is incomplete in oligopolistic markets, where expanded demand may be accompanied by changes in prices, plan offerings, or market structure. I use the event study to document these reduced-form movements and the structural model to evaluate the demand and entry channels. A full structural decomposition of provider price adjustment is left for future work.

3.7 Robustness of the Reduced-Form Evidence

The primary threat to identification in the ACP event study is that states with higher poverty rates (and hence higher ACP eligibility) were on differential broadband trends before the program, driven by factors unrelated to ACP itself. The inclusion of state-specific linear trends largely addresses this concern, and the pre-period evidence for most outcomes is consistent with parallel trends conditional on these controls.

Several additional checks support the baseline findings. First, the pre-period coefficients in Figure 3 are small and statistically insignificant for the outcomes that show the largest post-period effects (log high-speed price, plan shares), suggesting the treatment and control groups were on parallel trajectories before the program. Second, the BEAD results, which use a similar two-way fixed-effects specification, show clean pre-trends and similar qualitative patterns, providing an independent corroboration of the demand-side evidence. Third, Appendix F reports static DiD estimates for additional specifications; the qualitative conclusions are stable across these specifications.

¹⁰FCC/USAC Lifeline Program Data— annual disbursement reports explicitly frame outcomes as cost-per-subscriber (e.g., \$10/month/subscriber in 2015); available at usac.org/lifeline/resources/program-data.

A secondary concern is heterogeneity in treatment timing: because ACP’s eligibility variation is continuous and time-invariant, the standard two-way FE estimator is not subject to the staggered-adoption issues that complicate binary treatment designs [Sun and Abraham, 2021, Callaway and Sant’Anna, 2021]. The continuous-treatment design with event study coefficients is closer in spirit to a distributed-lag regression, and the interpretation of $\hat{\beta}_\tau^P$ as the dynamic causal response is standard in this context [Goldsmith-Pinkham et al., 2020].

4 Structural Model

4.1 Model Setup and Timing

Markets and products. Markets are indexed by t and firms (or providers) by f . A *market* is a U.S. census tract in 2020, before the implementation of the major IIJA broadband programs.¹¹ A *product position* is a provider-speed-tier pair (f, g, t) , where $g \in \{H, L\}$ denotes the IIJA high-speed tier ($\geq 25/3$ Mbps) and low-speed tier ($< 25/3$ Mbps). This classification captures a coarse but economically meaningful distinction in service quality relevant for household adoption decisions, distinguishing connections that can reliably support high-bandwidth activities from more limited service options. The indicator Y_{fgt} equals one if provider f offers tier g in tract t and zero otherwise; $Y_t = \{Y_{fgt}\}_{(f,g) \in \mathcal{J}_t^0}$ summarizes the product configuration. Provider entry in this model is *product positioning*: a provider that already serves some tracts in a county expands its product line by activating a tier in an additional tract or offering a second speed tier in an existing tract. The model thus captures both geographic expansion and product-line decisions within the local market.

The *outside option* is no fixed-broadband subscription. Market size M_t is the number of households in the tract, and the market share of the outside option is $s_{0t} = 1 - \sum_{j \in \mathcal{J}_t} s_{jt}$.

Timing and information. The following two assumptions formalize the timing and the information structure.¹²

Assumption 2 (Timing). *The game proceeds in two stages:*

1. *Product-positioning stage. Firms observe demand shifters X_t , cost shifters W_t , rivals’ observed footprints, and their own private fixed-cost shocks ζ_{fgt} . Based on this information,*

¹¹I use census-tract data and pre-2021 FCC Form 477 deployment data to ensure the baseline predates IIJA programs. This timing is important for the counterfactuals: simulated price discounts and fixed-cost reductions are applied to markets that do not already reflect ACP or BEAD effects.

¹²The timing assumption has been used in several studies. Examples include Fan and Yang [2024] for the brewery industry, Eizenberg [2014] in the personal computer industry, and Hidalgo [2024] for the broadband industry.

each firm simultaneously chooses its portfolio $Y_{ft} \in \mathcal{Y}_{ft}$.

2. *Pricing stage.* Conditional on the realized product configuration Y_t , firms observe demand shocks ξ_{jt} and marginal-cost shocks ω_{jt} and simultaneously set monthly plan prices in a complete-information Bertrand-Nash game.

Assumption 3 (Information). *Firms.* At the product-positioning stage, fixed-cost shocks ζ_{fgt} , market characteristics (X_t, W_t) , and rivals' potential market footprints are all common knowledge. The demand and cost shocks (ξ, ω) that will be realized in the second-stage pricing game are not yet known; only their joint distribution is common knowledge. Firms therefore compute expected variable profits by integrating over (ξ, ω) .

Econometrician. The shocks (ξ, ω, ζ) are unobserved. Market characteristics (X_t, W_t) and the set of observed product choices $\{Y_{fgt}\}$ are observed.

The separation between portfolio choice and pricing is standard in the entry and product-positioning literature [Mazzeo, 2002, Seim, 2006, Fan, 2013] and reflects the different time horizons involved: infrastructure deployment decisions are made months or years before price lists are set. The complete-information assumption on fixed costs is both analytically convenient and economically natural, as ISP deployment costs depend on local permitting, right-of-way negotiations, and contractor availability that are broadly understood within local infrastructure markets. Likewise, price equilibria are modeled under complete information because prices are publicly advertised and routinely monitored by competing firms.

Firm f 's first-stage objective is

$$\max_{Y_{ft} \in \mathcal{Y}_{ft}} \mathbb{E}_{\xi, \omega} \left[r_t^f(Y_{ft}, Y_{-ft}) \mid I_{ft} \right] - \sum_{g \in \{H, L\}} Y_{fgt} FC_{fgt}, \quad (4.1)$$

where $Y_{ft} = (Y_{fHt}, Y_{fLt})$, Y_{-ft} denotes rivals' portfolios, and $r_t^f(\cdot)$ is variable profit from the pricing game. Because a provider can offer at most two tiers in a tract, $\mathcal{Y}_{ft}$ contains four portfolios: no product, high-speed only, low-speed only, and both tiers.

The empirical difficulty is that Y_t is high dimensional. With roughly ten potential provider-tier positions in the average tract, a full solution of the entry game would require enumerating hundreds or thousands of product configurations per market. I therefore estimate first-stage fixed costs using the scalable discrete-game method of Fan and Yang [2024]. The key idea is to avoid solving the game during estimation. Instead, the model uses bounds on the incremental profit from offering each product position, which imply moment inequalities for the fixed-cost parameters. Once fixed costs are disciplined by these inequalities, counterfactual policies are evaluated by solving a best-response game under the new policy environment.

4.2 Consumer Demand

Household h 's indirect utility from product j in market t is

$$U_{hjt} = \delta_{jt} + \nu_{hg} + (1 - \rho)\varepsilon_{hjt}, \quad (4.2)$$

where δ_{jt} is mean utility, ν_{hg} is a group-level taste shock, and ε_{hjt} is Type-I extreme value.¹³ Mean utility is

$$\delta_{jt} = X'_{jt}\beta + \alpha p_{jt} + \xi_{jt}. \quad (4.3)$$

The vector X_{jt} includes download speed, upload speed, their squares, and a high-speed tier indicator. The parameter α is the marginal utility of price and ξ_{jt} is unobserved quality.

I use a single inside nest: all fixed-broadband products compete against the outside option of no fixed-broadband subscription. The single-nest specification is chosen for two reasons. First, it maintains a consistent nesting structure across all 69,085 census tracts. A two-level nest based on speed tier (e.g., high-speed products in one nest, low-speed in another) would be degenerate in markets where only one tier is offered: the within-nest share equals one, and the second-level nest collapses to a standard logit, producing an inconsistent likelihood across markets. Because a substantial fraction of rural tracts in the sample have only low-speed products, a tier-based two-level nest is not feasible without restricting the sample or treating single-tier markets separately. Second, the single-nest structure delivers a unique closed-form Bertrand-Nash pricing solution, which is essential for the entry counterfactuals: at each iteration of the product-positioning game, equilibrium prices must be computed rapidly across tens of thousands of markets and fixed-cost draws. A richer nesting structure (random coefficients, two-level nest) may admit multiple pricing equilibria in multiproduct settings [Nocke and Schutz, 2018], complicating the entry-game solution. The single-nest model therefore reflects a deliberate tractability choice, not a belief that the two-level structure is substantively incorrect. The Berry inversion [Berry, 1994] gives

$$\log(s_{jt}) - \log(s_{0t}) = X'_{jt}\beta + \alpha p_{jt} + \rho \log(s_{j|g,t}) + \xi_{jt}, \quad (4.4)$$

where $s_{j|g,t} = s_{jt}/(1 - s_{0t})$. In the tract-level data, the left-hand side is the CSS mean utility recovered from product-level implied shares.

¹³This paper departs from Fan and Yang [2024], who employ a random-coefficients demand model. Although such models allow for richer substitution patterns, they may generate non-unique pricing equilibria in multiproduct settings [Nocke and Schutz, 2018], thus complicating welfare and entry counterfactuals. Therefore, I use the nested logit model, which delivers a unique closed-form pricing solution and ensures a stable mapping from demand primitives to prices (e.g., Berry [1994]). Tractability is essential for evaluating firms' endogenous product offerings and entry decisions. I nonetheless rely on Fan and Yang [2024]'s framework to recover fixed-cost parameters in the entry stage.

Price and the within-inside share are endogenous. I instrument for these variables using BLP-style rivals' characteristics [Berry et al., 1995]:

$$z_{jt}^{(k)} = \frac{1}{J_t - 1} \sum_{\ell \neq j} X_{\ell t}^{(k)}, \quad k \in \{\text{downstream, upstream, Tier H}\}, \quad (4.5)$$

and the number of same-tier rivals [Aguirregabiria et al., 2024].

The logic of these instruments is standard [Berry et al., 1995, Nevo, 2001, Conlon and Gortmaker, 2020]. Rivals' speed characteristics affect a firm's own price only through strategic interaction: a firm with slower competitors can charge higher prices without losing as many subscribers. These characteristics are plausibly exogenous to a given firm's unobserved quality shock ξ_{jt} if firms choose deployment locations for reasons uncorrelated with rivals' cost shocks. The number of same-tier rivals captures the direct competitive pressure within the speed tier, which affects pricing but is not itself a choice variable of the focal firm in any given tract.

The preferred specification absorbs State and Provider $\times$ Tier fixed effects. State fixed effects absorb time-invariant state-level quality differences in broadband markets. Provider $\times$ Tier fixed effects absorb all provider-tier-specific mean quality differences, eliminating any systematic differences between, say, fiber products versus DSL products or between large national ISPs and small regional ones. Identification of $\hat{\alpha}$ and $\hat{\rho}$ in this specification comes from within-provider-tier variation across tracts in relative prices and market shares, conditional on state-level and provider-tier-level averages.

4.3 Oligopoly Price Competition

Assume that firm f 's variable profit in market t is

$$r_t^f = \sum_{j \in \mathcal{J}_t^f} (p_{jt} - mc_{jt}) M_t s_j(p_t, X_t, \xi_t), \quad (4.6)$$

where M_t is market size. The Bertrand-Nash first-order condition for product j is

$$s_{jt} + \sum_{k \in \mathcal{J}_t^f} (p_{kt} - mc_{kt}) \frac{\partial s_{kt}}{\partial p_{jt}} = 0. \quad (4.7)$$

Stacking conditions across products yields

$$\hat{\mu}^{BN} \equiv p - \hat{m}c = -H^{-1}s, \quad \text{where } H_{jk} = (\partial s_k / \partial p_j) * \Omega_{jr} \quad (4.8)$$

$$\Omega_{jr} = \begin{cases} 1 & \text{if } \exists f : r, j \in \mathcal{J}_i^f \\ 0 & \text{otherwise,} \end{cases}$$

I recover markups and marginal costs tract by tract and then estimate

$$\log(mc_{jt}) = W'_{jt}\gamma + \omega_{jt}, \quad (4.9)$$

where W_{jt} contains speed characteristics and the high-speed tier indicator.

4.4 Product Positioning and Fixed Costs

The supply-side object is a product-positioning decision. A product position is a provider $\times$ speed-tier pair in a census tract. Let f index canonical providers, $g \in \{H, L\}$ index speed tiers, and t index census tracts. The primitive decision is

$$Y_{fgt} = \mathbf{1}\{\text{provider } f \text{ offers speed tier } g \text{ in tract } t\}. \quad (4.10)$$

This definition treats entry as product positioning rather than entry into the national broadband industry. It captures both geographic expansion into a tract and product-line expansion within a tract. For example, a provider that already offers the low-speed tier in tract t enters the high-speed position if it also begins offering the high-speed tier.

Potential product positions. The observed data are organized at the tract $\times$ canonical-provider $\times$ speed-tier level, so each observed row corresponds to $Y_{fgt} = 1$. To estimate fixed costs, I also need feasible positions that are not chosen. I define the potential set using local provider footprints rather than assuming every national provider can enter every tract. Let $c(t)$ denote the county containing tract t . The baseline potential set is

$$\mathcal{J}_t^0 = \{(f, g) : \exists t' \in c(t) \text{ such that } Y_{fgt'} = 1\}. \quad (4.11)$$

Thus, provider-tier position (f, g) is feasible in tract t if that same provider-tier pair is observed somewhere else in the county. This restriction is economically natural: broadband expansion depends on nearby network presence, local permitting, and rights of way. It also keeps the potential choice set local enough to avoid treating providers with no plausible footprint as potential entrants.

For every feasible but unobserved position in $\mathcal{J}_t^0$, I impute product characteristics using predetermined information: provider-tier-state means, then provider-tier national means, then tier means. Observed positions retain their observed characteristics.

Table 5 reports the implied dimensions. The average tract contains 3.20 observed products but 10.54 potential provider-tier positions. A full market-level portfolio model would require evaluating $2^{|\mathcal{J}_t^0|}$ configurations per market (up to thousands at the 75th percentile), motivating the Fan and Yang [2024] approach below.

Table 5: Dimension of Observed and Potential Product Sets

Product-set definition	Mean	P25	Median	P75	P90
Observed products in tract, J_t^{obs}	3.20	2	3	4	5
Observed providers in tract	2.36	2	2	3	4
County-footprint positions, $ \mathcal{J}_t^0 $	10.54	6	9	13	21
County providers	8.13	4	7	10	16

Notes: The unit of observation is a census tract. Observed products are provider-tier products present in the tract. The county-footprint potential set includes provider-tier positions observed somewhere in the same county as the tract.

4.5 Multiple Equilibria and Equilibrium Selection

The product-positioning game generically admits multiple Nash equilibria: for a given draw of fixed costs, there may be several product configurations that are mutually best-response stable. This is a fundamental challenge in empirical models of entry, recognized since Bresnahan and Reiss [1991] and Tamer [2003], and addressed directly by Ciliberto and Tamer [2009] and Fan and Yang [2024].

The estimation stage does not require equilibrium selection because the moment-inequality approach only requires that the observed product configuration is *not dominated*. Every Nash equilibrium satisfies this condition, so the bounds on fixed costs are valid regardless of how equilibrium is selected in practice. This is precisely the advantage of the Fan and Yang [2024] method over likelihood-based approaches that require a unique equilibrium assumption.

The counterfactual stage does require solving for post-policy equilibria. I handle this as follows. For each market and each draw of fixed costs, I run iterated best-response from three starting points: the observed pre-policy configuration, the empty configuration, and the full feasible configuration.¹⁴ The algorithm is declared converged when no firm can profitably deviate from the current portfolio, checked by a firm-by-firm Nash verification. If multiple fixed points are found from different starting configurations, I retain all of them. The reported national-level bounds combine equilibrium multiplicity and fixed-cost parameter uncertainty by stacking market-level lower and upper outcomes across equilibria and fixed-cost draws and reporting the 2.5th and 97.5th percentile bounds.

¹⁴Fan and Yang [2024] consider two starting points: the empty and the full feasible configuration.

Markets where no fixed point is found within the iteration limit (fewer than one percent of markets) are assigned the pre-policy configuration. This conservative fallback ensures the bounds remain valid; it is likely to understate the response of hard-to-solve markets. Extensive convergence diagnostics, reported in Appendix J, show that the algorithm converges in the vast majority of markets, and the distribution of convergence failures is not systematically correlated with market size or policy intensity.

4.6 Measurement and Limitations

The structural analysis rests on several measurement assumptions. Appendix G discusses these in detail. The three main concerns are: (i) product-level market shares are inferred rather than directly observed, using the CSS procedure of Crawford et al. [2019]; (ii) prices are imputed from URS advertised plan data rather than actual transaction prices; and (iii) the model is static with a December 2020 baseline, so it cannot be directly validated against post-program outcomes. Robustness checks described in Appendix G support the main quantitative conclusions despite these limitations.

5 Estimation Results

I estimate the model in three steps. First, I estimate the nested-logit demand system using the Berry inversion in (4.4). The preferred specification instruments for price and the within-inside share using rivals’ product characteristics and the number of same-tier rivals, and absorbs State and Provider $\times$ Tier fixed effects. Second, I use the demand estimates and the Bertrand-Nash first-order conditions in (4.7) to recover markups and marginal costs tract by tract. Third, I estimate fixed costs from the product-positioning inequalities implied by the first-stage model. The moment inequalities follow Fan and Yang [2024]: observed product choices are required not to be dominated, and the resulting fixed-cost parameters are reported as bounds. Appendix J gives the step-by-step construction of the fixed-cost moments, the weight matrix, and the confidence region.

5.1 Consumer Demand

Table 6 reports four demand specifications. Columns (1) and (2) are OLS; columns (3) and (4) instrument for price and the within-inside share. Even-numbered columns absorb State and Provider $\times$ Tier fixed effects. Column (4) is the preferred specification.

The preferred price coefficient is negative and economically large: IV amplifies its magnitude by a factor of 36 $\times$ relative to OLS ($-0.26 \rightarrow -9.42$), consistent with substantial

Table 6: Nested Logit Demand Estimates

	OLS		2SLS (BLP instruments)	
	(1)	(2)	(3)	(4)
Price / \$100 [$\hat{\alpha}$]	−0.260*** (0.002)	−0.329*** (0.004)	0.818*** (0.005)	−9.424*** (0.696)
Within-share [$\hat{\rho}$]	0.245*** (0.001)	0.214*** (0.001)	0.255*** (0.002)	0.220*** (0.008)
Download speed (Gbps)	0.308*** (0.003)	0.367*** (0.003)	0.139*** (0.010)	0.435*** (0.015)
Download speed ²	−0.098*** (0.002)	−0.107*** (0.002)	−0.079*** (0.009)	−0.152*** (0.004)
Upload speed (Gbps)	0.008*** (0.003)	0.017*** (0.003)	0.248*** (0.009)	0.021* (0.011)
Upload speed ²	0.047*** (0.002)	0.051*** (0.002)	0.022** (0.009)	0.089*** (0.003)
High-speed indicator	0.122*** (0.001)	n.s. —	−0.088*** (0.002)	n.s. —
N	220,821	220,821	220,821	220,821
R^2	0.782	—	—	—
State FEs	No	Yes	No	Yes
Provider×Tier FEs	No	Yes	No	Yes
Mean own-price elasticity	−0.18	−0.22	0.57	−6.33

Notes: HC1 robust standard errors in parentheses for OLS; heteroskedasticity-robust for 2SLS. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. n.s. = not significant; the high-speed indicator is collinear with Provider×Tier FEs in columns (2) and (4) and is omitted. BLP instruments: mean rivals' downstream speed, upstream speed, tier-H share, and number of same-tier rivals (all tract-level). Column (4) is the preferred specification.

positive correlation between prices and unobserved quality. This amplification reflects the fact that URS prices are imputed from plan-level data and contain measurement error that attenuates OLS, while the BLP instruments correct for both classical measurement error and the endogeneity of prices with respect to unobserved product quality.

The nesting parameter $\hat{\rho} = 0.220$ indicates moderate within-group correlation among fixed-broadband products, significantly below the estimate from the state-level specification ($\hat{\rho} = 0.646$). The difference reflects the different levels of aggregation: at the tract level, product availability varies sharply across providers, reducing within-nest correlation. Download speed enters positively and concavely, consistent with diminishing marginal utility from very high

speeds.

The relatively low $\hat{\rho}$ implies substitution patterns that are closer to proportional (multinomial logit) than to perfect within-nest substitution. This is a known limitation of the single-nest specification: with $\hat{\rho}$ near zero, the model does not generate strong cross-tier substitution asymmetries. A two-level nest (speed tier within technology, or technology within speed tier) would allow high-speed and low-speed products to substitute more strongly within tier. As discussed above, however, such a structure is not feasible here because many rural tracts contain only one speed tier, causing the second nest to be degenerate. The tractability requirements of the entry-game counterfactuals further preclude random-coefficients demand, which may generate non-unique pricing equilibria [Nocke and Schutz, 2018]. Given these constraints, $\hat{\rho} = 0.220$ should be interpreted as capturing residual within-broadband correlation after absorbing provider-tier and state fixed effects, not as a claim that all broadband products are equally close substitutes.

The mean own-price elasticity in the preferred specification is -6.33 :

$$\varepsilon_{jj} = \frac{\hat{\alpha}}{100} p_{jt} \left[\frac{1}{1 - \hat{\rho}} - \frac{\hat{\rho}}{1 - \hat{\rho}} s_{j|g,t} - s_{jt} \right]. \quad (5.1)$$

Comparison with related estimates. The estimated own-price elasticity, $\hat{\varepsilon} = -6.33$, is somewhat larger in magnitude than the household- or market-level estimates commonly reported in the broadband demand literature, which often lie between approximately -1 and -4 [Rosston et al., 2010, Nevo et al., 2016]. However, it is close to several recent estimates from broadband and related subscription markets. For example, Kearns [2024] reports broadband price elasticities of -4.73 , -4.46 , and -4.16 for low-, middle-, and high-income households, respectively. Goetz [2019] estimates an average own-price elasticity of approximately -5.9 , while Hidalgo [2024] finds an average elasticity of -4.6 in the Colombian broadband market. Related evidence from the cable television industry is also similar in magnitude: Crawford et al. [2019] reports elasticities of -4.1 for cable services and -5.4 for satellite services.

Several features of the present setting help explain why the estimate is at the upper end of this range. First, demand is estimated at the product level, where a product is defined by a provider-tier-tract combination, rather than at the household or aggregate market level. The relevant margin is therefore substitution across broadband products within a local market, not the extensive-margin decision to subscribe to broadband at all. Product-level substitution elasticities are naturally larger in magnitude than market participation elasticities.

Second, the nested logit specification implies that the own-price elasticity depends directly on the product market share, s_{jt} . Evaluated at the sample mean share of 0.259 , the elasticity formula yields $\hat{\varepsilon} = -6.33$. Evaluating the same expression at lower shares, which are more

representative of competitive rural markets, produces elasticities closer to -3 to -5 .

Finally, an earlier version of this paper estimated demand using a state-by-year aggregation approach that constructed shares by combining ACS aggregate broadband demand with plan availability measures. That specification produced a mean own-price elasticity of approximately -5.87 , close to the current estimate. This provides reassurance that the tract-level share construction procedure does not mechanically generate the elasticity magnitude.

Overall, the estimated elasticity appears economically plausible for product-level broadband demand within local markets. While larger than many participation-margin estimates, it is closely aligned with recent broadband estimates and with evidence from related subscription industries.

Elasticity heterogeneity across market types. A key feature of the nested logit model is that own-price elasticities vary with market shares, as seen in (5.1). This heterogeneity has substantive implications for welfare calculations. In low-share markets (typically rural tracts with fewer products and higher outside-option shares), own-price elasticities are smaller in magnitude: at a product share of $s_{jt} = 0.10$ with $\hat{\rho} = 0.220$, the formula yields an elasticity of approximately -3.5 . In high-share markets (dense urban tracts with many products and low outside-option shares), own-price elasticities are larger: at $s_{jt} = 0.45$, the elasticity is approximately -8.7 . The mean of -6.33 reflects the distribution of market shares across tracts, which is skewed toward intermediate values.

This elasticity heterogeneity matters for the welfare effects of both programs. ACP raises consumer surplus most where elasticity is highest—where consumers are most price-sensitive and closest to the margin of subscribing—which tends to be in lower-income tracts with moderate to high baseline adoption. BEAD, which operates through fixed-cost relief rather than price changes, is less sensitive to demand elasticity and instead depends on the gap between variable profits and fixed costs. The combination of high-elasticity markets (well-served by ACP) and low-elasticity but high-fixed-cost markets (well-served by BEAD) reinforces the geographical complementarity between the two programs documented in Section 6.3.

Instrument strength. The BLP instruments are strongly predictive of both endogenous variables. Table 7 reports the excluded-instrument coefficients and weak-identification diagnostics for the preferred specification (column 4 of Table 6). Each first-stage regression also includes the full set of included exogenous controls and the fixed effects used in the second-stage demand specification.

Because the model contains two endogenous variables, the Angrist–Pischke diagnostics

should be interpreted as equation-specific tests of instrument relevance rather than as a complete weak-instrument test for the full endogenous regressor vector. The robust Angrist–Pischke Wald statistics are 435.7 for price and 167,617.0 for the within-inside share. The corresponding conditional first-stage statistics are 320.1 and 42,082.4, respectively. The Cragg–Donald minimum eigenvalue statistic is 623.2, providing additional system-level evidence against weak identification under homoskedasticity.

The signs are economically intuitive. Rivals with higher download speeds and larger high-speed tier shares are associated with lower prices for the focal product, consistent with competitive pressure, and with lower within-group shares, consistent with substitution toward rival products. A larger number of same-tier rivals similarly reduces both price and within-group share. The overidentification test rejects exact exclusion at conventional levels ($\chi^2 = 14.84$, $p = 0.001$), but with more than 216,000 observations the test has high power against small deviations from exact exclusion. I therefore interpret the instruments as strong and approximately valid rather than exactly excluded.

Table 7: First-Stage Diagnostics: Preferred 2SLS-FE Specification

	Price / \$100	$\log(s_{j g})$
Mean rivals' download speed (Gbps)	−0.006***	−0.014***
Mean rivals' upload speed (Gbps)	0.004***	0.036***
Mean rivals' high-speed share	−0.006***	−0.276***
Number of same-tier rivals	−0.002***	−0.257***
Included exogenous controls	Yes	Yes
State fixed effects	Yes	Yes
Provider-by-tier fixed effects	Yes	Yes
Angrist–Pischke Wald statistic	435.7	167,617.0
Shea partial R^2	0.0007	0.1599
Conditional first-stage F -statistic	320.1	42,082.4
Cragg–Donald minimum eigenvalue statistic	623.2	

Notes: The table reports coefficients on excluded instruments. Each first stage also includes the exogenous controls and fixed effects shown above.

Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.2 Marginal Costs and Markups

Table 8 reports the marginal-cost regressions. Column (1), which does not include fixed effects, is the preferred supply specification. This choice reflects the fact that much of the relevant cost variation comes from differences across providers, technologies, speed

tiers, and plan characteristics that would be absorbed or sharply attenuated by rich fixed effects. In specifications with fixed effects, the estimated signs of several cost shifters become economically counterintuitive and the implied marginal costs are negative for a larger share of observations. I therefore use the parsimonious specification in column (1) as the baseline for recovering marginal costs.

Table 8: Supply-Side OLS: Dependent Variable $\log(\hat{m}c_{jt})$

	(1) No FE Preferred	(2) Two-way FE
Constant	3.742*** (0.002)	—
Download speed (Gbps)	0.287*** (0.010)	−0.022*** (0.003)
Download speed ² (Gbps ²)	−0.026*** (0.009)	−0.003*** (0.001)
Upload speed (Gbps)	−0.412*** (0.009)	−0.003 (0.003)
Upload speed ² (Gbps ²)	0.039*** (0.009)	0.006*** (0.001)
High-speed indicator	0.341*** (0.003)	—
Observations	198,796	198,796
R^2	0.226	0.001
RMSE	0.394	0.173
State FE	No	Yes
Provider×Tier FE	No	Yes

Notes: Dependent variable is $\ln(\hat{m}c_{jt})$ where $\hat{m}c_{jt} = p_{jt} - \hat{\mu}_{jt}^{BN}$. Sample restricted to products with positive implied marginal costs. HC1 robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Tier H and the constant are collinear with Provider×Tier FEs in column (2) and are omitted.

The preferred specification implies that download speed raises marginal cost at a diminishing rate. High-speed tier products carry a marginal-cost premium of $\exp(0.341) - 1 \approx 41$ percent relative to low-speed tier products. Upload speed is negatively related to marginal cost in the cross-section, consistent with fiber-grade providers combining high upload capacity with newer, more efficient infrastructure. Of 220,821 products, 22,025 (9.97 percent) yield negative implied marginal costs, concentrated among multiproduct firms, a known feature of Bertrand-Nash markup recovery in differentiated-products markets [Nevo, 2001].

Conditional on positive marginal costs, the median Lerner index is 18.2 percent. This

figure implies a price-cost margin of roughly 18 cents per dollar of revenue for the typical broadband product. For single-product firms—for whom the Bertrand-Nash first-order condition simplifies to the inverse-elasticity rule—the mean Lerner index of 15.3 percent is close to the theoretical benchmark $1/|\hat{\epsilon}| = 15.8$ percent, providing an internal consistency check on the structural parameters. The higher Lerner indices for multi-product firms (mean 27.9 percent) reflect the internalization of cross-product cannibalization effects [Nevo, 2001].

Comparison with related estimates. The median Lerner index of 18.2 percent is plausible relative to estimates in related markets. Nevo [2001] finds Lerner indices of 16–45 percent for ready-to-eat cereals, a market characterized by high product differentiation and few national competitors. Crawford, Shcherbakov, and Shum [2019] find similar magnitude markups in cable television. The recovery of markups depends on the demand elasticity, and the 18.2 percent median is directly consistent with a mean own-price elasticity of -6.33 through the Lerner formula. Average marginal costs of approximately \$49 per month for all products are economically plausible given the capital intensity of broadband infrastructure. The constant of 3.742 in the log-marginal-cost regression implies $\exp(3.742) \approx \$42$ per month at zero speeds, with speed characteristics adding to this baseline; high-speed products average approximately \$57 per month in marginal costs.

Tables 9 and 10 report the full distribution of markups, marginal costs, and Lerner indices by speed tier and firm type.

Several patterns merit discussion. High-speed (Tier H) products have marginally lower Lerner indices than low-speed (Tier L) products (median 0.156 vs. 0.220). This is consistent with high-speed products facing greater effective competition: fiber and modern cable products occupy a more competitive landscape than legacy DSL. Tier L products, largely DSL, exhibit higher markups in many markets because they operate as de facto monopolists in low-density areas where no competing technology exists. The substantially elevated Lerner indices for multi-product firms (median 0.248 vs. 0.138 for single-product) reflect the well-understood mechanism in differentiated-products markets: firms with multiple products internalize the cannibalization effect between them, setting higher markups on each product than a single-product firm would [Nevo, 2001]. In broadband markets, this internalization means that an ISP offering both high-speed and low-speed tiers in the same tract prices both higher than it would if the tiers were offered by independent firms, reducing consumer surplus for buyers who would have benefited from intra-firm competition. This finding has a direct counterfactual implication: BEAD-induced entry, which typically introduces a new provider rather than a new product tier from an existing provider, generates competitive pressure that is absent when an incumbent simply adds a second tier.

Table 9: Bertrand-Nash Markup Distribution by Speed Tier

	All	Tier H	Tier L	$\hat{m}c > 0$ only
Observations	220,821	138,460	82,361	198,796
% negative $\hat{m}c$	9.97%	8.62%	12.25%	—
<i>Panel A: Markup $\hat{\mu}$ (\$/month)</i>				
Mean	23.03	21.62	25.40	—
Median	12.68	12.25	14.11	—
P90	35.37	35.37	106.11	—
<i>Panel B: Marginal Cost $\hat{m}c$ (\$/month)</i>				
Mean	48.78	57.32	34.41	—
Median	54.51	59.82	47.03	—
P90	83.13	87.73	60.77	—
<i>Panel C: Lerner Index $\hat{L} = \hat{\mu}/p$</i>				
Mean	—	0.188	0.262	0.215
Median	—	0.156	0.220	0.182
P90	—	0.302	0.452	0.357

Notes: Tier H: download ≥ 25 Mbps; Tier L: < 25 Mbps. Lerner index $= \hat{\mu}/p$; values exceeding 1 arise when implied MC is negative. The “ $\hat{m}c > 0$ only” column restricts to the 90.0% of products with non-negative implied marginal costs.

Table 10: Markup and Lerner Index by Firm Type

	Single-Product	Multi-Product
Observations	105,597	115,224
% negative $\hat{m}c$	4.76%	14.75%
<i>Panel A: Markup $\hat{\mu}$ (\$/month)</i>		
Mean	15.85	29.61
Median	10.86	16.80
P90	15.15	106.11
<i>Panel B: Marginal Cost $\hat{m}c$ (\$/month)</i>		
Mean	64.51	34.37
Median	64.85	48.77
<i>Panel C: Lerner Index $\hat{L} = \hat{\mu}/p$, $\hat{m}c > 0$ only</i>		
Mean	0.153	0.279
Median	0.138	0.248
P25	0.120	0.190
P75	0.174	0.323
P90	0.219	0.463

Notes: Single-product firms offer one tier in the tract; multi-product firms offer both tiers or operate in multiple tracts. Lerner index restricted to products with $\hat{m}c > 0$.

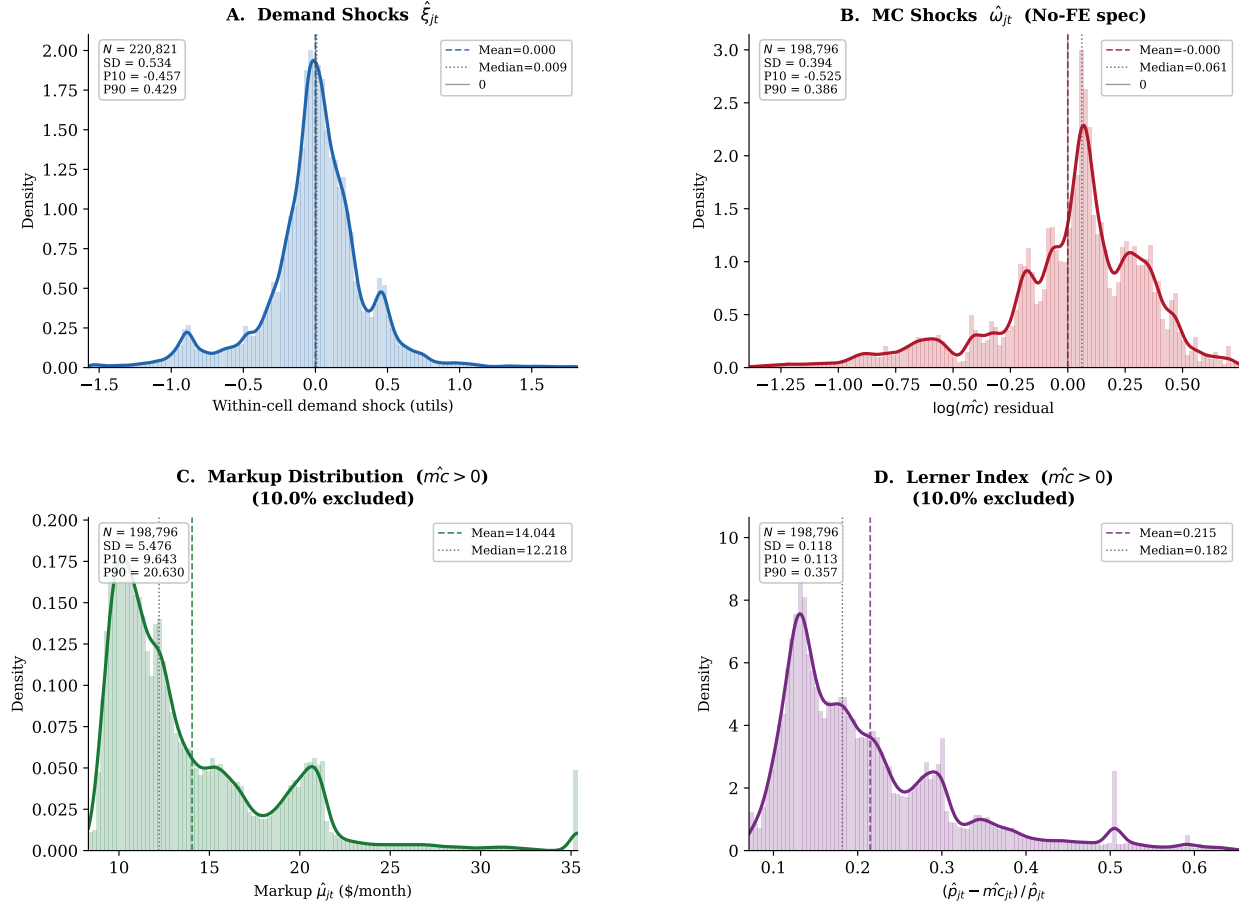


Figure 5: Empirical distributions of demand shocks $\hat{\xi}_{jt}$, marginal-cost shocks $\hat{\omega}_{jt}$, Bertrand-Nash markups, and price-to-marginal-cost ratios, trimmed to the 1st–99th percentile for display. Dashed lines mark means; dotted lines mark medians. The distribution of $\hat{\xi}_{jt}$ is approximately centered at zero and symmetric, consistent with the orthogonality conditions from the demand system. Markups are concentrated in the \$10–\$30 range with a modest right tail driven by multi-product firms. Price-cost ratios have median near 1.20, reflecting the 18.2 percent median Lerner index.

Figure 5 reports key structural diagnostics. Demand shocks are centered near zero and approximately symmetric, consistent with the orthogonality conditions imposed by the BLP instruments. This is an important internal validity check: if the instruments were correlated with unobserved quality, the residuals would be systematically shifted. Marginal-cost shocks are also approximately symmetric and centered near zero. The markup distribution is right-skewed, reflecting the higher markups of multi-product firms, and is consistent with the Lerner index distributions in Tables 9 and 10 above. Appendix I provides supplementary markup diagnostics including the marginal-cost distribution by technology type.

5.3 Fixed-Cost Estimates

I parameterize fixed costs by market-size cells and speed tier:

$$FC_{fgt} = \theta_{S(t),g} + \sigma_{\zeta}\zeta_{fgt}, \quad \zeta_{fgt} \sim N(0, 1), \quad (5.2)$$

where $S(t) \in \{\text{Small, Medium, Large}\}$ denotes the market-size tertile of tract t and $g \in \{H, L\}$ is the speed tier. This specification allows fixed costs to differ between small, medium, and large markets and between high-speed and low-speed positions, while $\sigma_{\zeta}\zeta_{fgt}$ captures private position-level cost heterogeneity.

For each position, I compute lower and upper bounds on the incremental variable profit from offering the product. The logic is intuitive: if a provider *does* offer a tier, the fixed cost must be low enough that keeping the position is better than dropping it (lower bound on variable profit $\Rightarrow$ upper bound on fixed cost is not binding). If a provider *does not* offer a feasible tier, the fixed cost must be high enough that entering is not dominant (upper bound on variable profit $\Rightarrow$ lower bound on fixed cost). These dominance restrictions generate moment inequalities that bound the fixed-cost parameters without requiring equilibrium selection [Fan and Yang, 2024, Ciliberto and Tamer, 2009, Tamer, 2003].

Figure 6 summarizes the identifying logic. Further sanity checks are available in Appendix K. Table 11 reports the resulting bounds.

Table 11: Fixed-Cost Parameter Bounds by Market Size and Speed Tier

Market size	Tier	Bound (θ)	SD bound (σ_{ζ})
Small	High speed	[717, 9,790]	[1,734, 5,068]
	Low speed	[532, 8,572]	[886, 3,864]
Medium	High speed	[1,197, 15,773]	[3,191, 7,875]
	Low speed	[900, 13,862]	[1,578, 5,081]
Large	High speed	[1,930, 24,579]	[5,307, 14,871]
	Low speed	[1,433, 21,895]	[2,803, 10,887]

Notes: Bounds are reported in monthly dollars. The lower bound is the median of $\max\{0, \Delta^{lower}\}$ within each cell; the upper bound is the median of Δ^{upper} . “SD bound” reports the corresponding within-cell standard deviations.

Economic interpretation and literature comparison. The fixed-cost bounds are reported in monthly dollars per product position. Monthly fixed costs are best interpreted as the flow equivalent of the annualized capital and operational expenditures required to maintain a broadband product in a census tract: network maintenance, equipment leasing, local permitting, and customer support. Converting the monthly bounds to annualized costs,

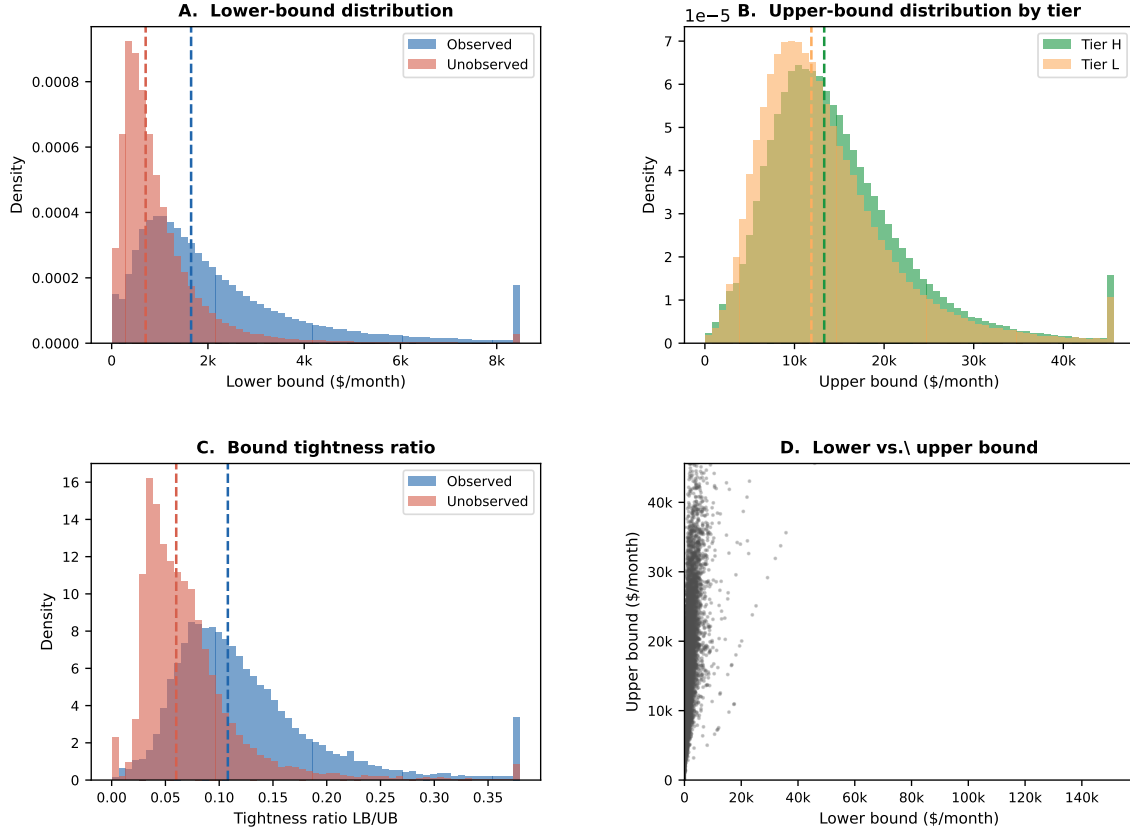


Figure 6: Fan–Yang fixed-cost bounds from product-positioning decisions. Each panel illustrates the dominance restrictions that bind for an observed entry or non-entry: observed products generate upper bounds on fixed costs (the firm’s variable profit exceeds the fixed cost); unobserved feasible products generate lower bounds (variable profit does not cover the fixed cost at any equilibrium). The moment inequalities aggregate these restrictions across positions to produce identified bounds on the six fixed-cost parameters.

the estimated range for small-market high-speed products is approximately \$8,600–\$117,500 per year, and for large-market high-speed products approximately \$23,200–\$295,000 per year. These figures are broadly consistent with industry estimates of last-mile fiber deployment costs, which range from roughly \$2,000 to \$30,000 per household passed in rural areas, depending on terrain and density. A tract with 1,679 average households (the sample mean) multiplied by a cost of \$2,000–\$30,000 per household implies annualized capital costs in the range of roughly \$100,000–\$3,000,000, broadly overlapping with the upper range of my structural bounds. The wide bounds reflect both genuine cost heterogeneity and partial identification: the moment inequalities do not point-identify fixed costs, only bound them.

To put these numbers in perspective, the median lower bound on variable profit for observed positions is approximately \$1,650 per month (from Appendix K), while the median upper bound is approximately \$15,318. For fixed costs in the middle of the estimated range, roughly \$3,000–\$8,000 per month, deployment breaks even only in tracts with substantial market presence. This is consistent with the well-documented finding that broadband deployment in low-density rural areas is privately unprofitable at cost levels typical of urban markets [Kearns, 2024, Espín and Rojas, 2024]. The increasing fixed costs from small to large markets reflect the fact that large tracts offer higher absolute variable profits, which raises the threshold at which marginal deployment becomes privately attractive.

The connection between fixed-cost bounds and BEAD counterfactuals is direct. A BEAD subsidy of τ percent reduces the private fixed-cost burden to $(1 - \tau)FC$. For positions whose unsubsidized fixed cost falls between the variable profit lower and upper bounds—the “contestable margin”—a subsidy can flip the deployment decision from non-entry to entry. The width of the fixed-cost bounds reported in Table 11 therefore also determines the set of positions that become potentially deployable under a given subsidy rate. For small-market high-speed products, a 75 percent subsidy reduces the effective annual fixed cost to roughly \$2,150–\$29,370, which is substantially below the median upper bound on variable profits for such positions. This calculation motivates the large entry responses documented in the counterfactual simulations at high subsidy rates, and shows that the $\sim 341,000$ maximum new entries at the 75 percent rate are consistent with the fixed-cost parameters.

6 Counterfactual Policy Analysis

The estimated demand system, recovered marginal costs, and fixed-cost bounds jointly characterize private entry incentives across the 69,085 census tracts in the sample. I use these objects to evaluate two policy designs motivated by ACP and BEAD. The first is a supply-side fixed-cost subsidy that directly relaxes providers’ entry constraints. The second

is a demand-side price subsidy that lowers the price paid by consumers while reimbursing firms at the gross price. The policies move different primitives: BEAD reduces deployment fixed costs, while ACP reduces the effective consumer price. That distinction is what drives the welfare comparison below.

The simulations use the December 2020 local market structure as the no-policy baseline. Because this baseline predates ACP and BEAD implementation, the counterfactual discounts and fixed-cost subsidies are applied to an environment that has not already incorporated the effects of those programs. The counterfactuals are solved market by market. For each draw of fixed costs, firms choose product portfolios rather than isolated products. I solve the post-policy game by iterated firm best responses, starting from the observed configuration, the empty configuration, and the all-feasible configuration (see Appendix J). Candidate outcomes are retained only when they pass a firm-by-firm Nash best-response check. When several fixed points are found, I report the range of outcomes across them. The reported national bounds combine equilibrium multiplicity with fixed-cost uncertainty by summing market-level lower and upper outcomes for each draw and then reporting the 2.5th percentile of lower outcomes and the 97.5th percentile of upper outcomes across draws.

6.1 Counterfactual Policy Designs

6.1.1 Broadband Equity, Access, and Deployment Program

BEAD is a supply-side program that subsidizes broadband deployment in unserved and underserved areas. The program can cover up to 75 percent of eligible project costs, with a minimum 25 percent match from the subgrantee.¹⁵ I implement this as a reduction in the private fixed cost of offering a product. For each fixed-cost draw r and subsidy rate $\tau \in \{0.25, 0.50, 0.75\}$:

$$FC_j^{cf,r}(\tau) = (1 - \tau)FC_j^r. \quad (6.1)$$

The policy does not directly change prices or marginal costs. Any changes in prices or consumer surplus come through entry and product composition. Government spending is

$$G^{BEAD,r}(\tau) = \frac{\tau}{1 - \tau} \text{PFC}^r(\tau), \quad (6.2)$$

where $\text{PFC}^r(\tau)$ is the post-subsidy private fixed cost paid by firms. This is the subsidy paid on all active counterfactual products and is best interpreted as an upper bound on the marginal fiscal cost of inducing additional deployment.

¹⁵See <https://broadbandusa.ntia.doc.gov/>.

6.1.2 Affordable Connectivity Program

ACP is a demand-side program. Eligible households receive a monthly discount, and participating providers are reimbursed for the discount. For discounts $a \in \{10, 20, 30\}$ dollars, the consumer-facing price is

$$p_j^c(a) = \max\{p_j^g - a, 0.01\}, \quad FC_j^{cf,r} = FC_j^r, \quad (6.3)$$

where the gross reimbursed price p_j^g , marginal cost, and product fixed cost are unchanged. The policy lowers the price relevant for household choice but leaves the firm's per-subscriber margin equal to $p_j^g - mc_j$.

The implementation uses universal eligibility: every household in every tract is treated as eligible for the discount. This isolates the price-subsidy mechanism and avoids introducing additional measurement error from incomplete eligibility data. Under this assumption, government outlay is

$$G^r(a) = a Q^{elig,r}(a). \quad (6.4)$$

In this implementation, ACP works through the consumer-facing price. Gross ISP prices are held fixed, so the counterfactual does not model how providers might adjust posted prices after the demand shock. Those adjustments are outside the structural ACP exercise.

6.2 Counterfactual Results

6.2.1 Broadband Equity, Access, and Deployment Program

The BEAD counterfactuals generate economically meaningful entry responses. A 25 percent fixed-cost reduction raises consumer surplus from a baseline bound of \$2.18–\$2.65 billion per month to \$2.49–\$2.87 billion. At a 75 percent fixed-cost reduction, consumer surplus reaches \$2.86–\$3.36 billion per month. The corresponding social-welfare bound rises from \$2.69–\$4.06 billion in the baseline to \$3.85–\$4.46 billion at the largest subsidy. These gains are accompanied by substantial expansion in product availability: the 75 percent subsidy adds between 115,237 and 340,967 provider-tract entries, with most of the response coming from high-speed products.

The welfare decomposition clarifies the economic mechanism. Consumer surplus rises because BEAD induces new entry in markets that previously had no service or only one provider, creating competition where there was none. Producer surplus also rises on net because new entrants earn positive variable profits—otherwise they would not enter—while incumbents face slightly more competition. Social welfare increases because the new products

Table 12: BEAD Counterfactual Outcomes

	Baseline	25% FC reduction	50% FC reduction	75% FC reduction
Consumer surplus (\$B/month)	[2.18, 2.65]	[2.49, 2.87]	[2.60, 3.15]	[2.86, 3.36]
Producer surplus (\$B/month)	[0.51, 1.41]	[0.73, 1.06]	[0.82, 0.96]	[0.99, 1.10]
Social welfare (\$B/month)	[2.69, 4.06]	[3.23, 3.93]	[3.43, 4.11]	[3.85, 4.46]
Average price (\$/month)	[66.30, 78.38]	[70.38, 73.95]	[71.45, 72.93]	[71.99, 72.36]
Average market share	[0.753, 0.825]	[0.796, 0.846]	[0.813, 0.871]	[0.846, 0.885]
New firm-markets	[0, 78,716]	[32,721, 129,744]	[53,132, 232,388]	[115,237, 340,967]
New high-speed products	[0, 63,875]	[22,278, 115,772]	[37,712, 200,602]	[88,180, 283,736]
New low-speed products	[0, 25,127]	[11,525, 37,002]	[18,469, 64,791]	[38,540, 133,586]

Notes: Bounds are $[Q_{2.5}(\underline{Y}^r), Q_{97.5}(\bar{Y}^r)]$ across 50 fixed-cost draws. Welfare in monthly billions of dollars. Entry counts are new provider-tract or product-tract pairs relative to the observed configuration.

generate surplus that exceeds their fixed costs from the private perspective (the subsidy covers that gap). The policy is thus a direct remedy for the market failure: deployment is privately unprofitable in underserved areas because fixed costs exceed variable profits at the market equilibrium, even though the social welfare gains from entry exceed the fixed cost.

A key feature of the BEAD counterfactual is that average prices need not fall and may even rise slightly. This is a composition effect, not a failure. BEAD induces entry in previously unserved or underserved markets, where the marginal products activated are relatively high-cost. These products enter at prices above the existing market average. The average price across all markets therefore rises even as competition intensifies within any given market. Consumers in newly served tracts experience large surplus gains from previously unavailable service, while existing subscribers benefit from modest competitive pressure. Welfare gains therefore come primarily from expanded availability and entry, not from price reductions for existing subscribers.

The entry response is nonlinear in the subsidy rate. Moving from a 25 percent to a 50 percent fixed-cost reduction roughly doubles the upper bound on new firm-market entries; moving to a 75 percent reduction expands entry much more sharply. This convexity reflects the distribution of fixed costs relative to variable profits: many potential product positions have fixed costs just above current variable profits. Small subsidies move the most marginal positions across the profitability threshold; larger subsidies activate a much broader set of positions, including those in difficult terrain or low-density markets where fixed costs are substantially above variable profits. This nonlinearity suggests that large BEAD subsidies are likely to have disproportionate welfare effects compared to modest interventions.

An important complementarity between BEAD’s entry channel and ACP’s demand channel is visible in the counterfactuals. ACP raises variable profits by expanding demand: at the \$30

discount, the average market share rises from 0.75–0.83 to 0.92–0.97. Higher variable profits, in turn, raise the internal return to entry for providers already near the margin of deployment. In a dynamic model, ACP’s demand expansion would therefore make some marginal BEAD positions more attractive. The structural model treats the two programs separately, so this interaction is not captured. The table in Section 6.3 accounts for this by noting that ACP also induces modest entry (18,000–28,000 new firm-markets at the \$30 discount), but the main mechanism is demand expansion rather than entry. A natural extension would simulate the two programs simultaneously, allowing the demand expansion from ACP to interact with the fixed-cost reduction from BEAD. Based on the separate simulations, the two channels appear to reinforce rather than offset each other, suggesting that the combined welfare effects of simultaneous implementation would exceed the sum of the individual effects.

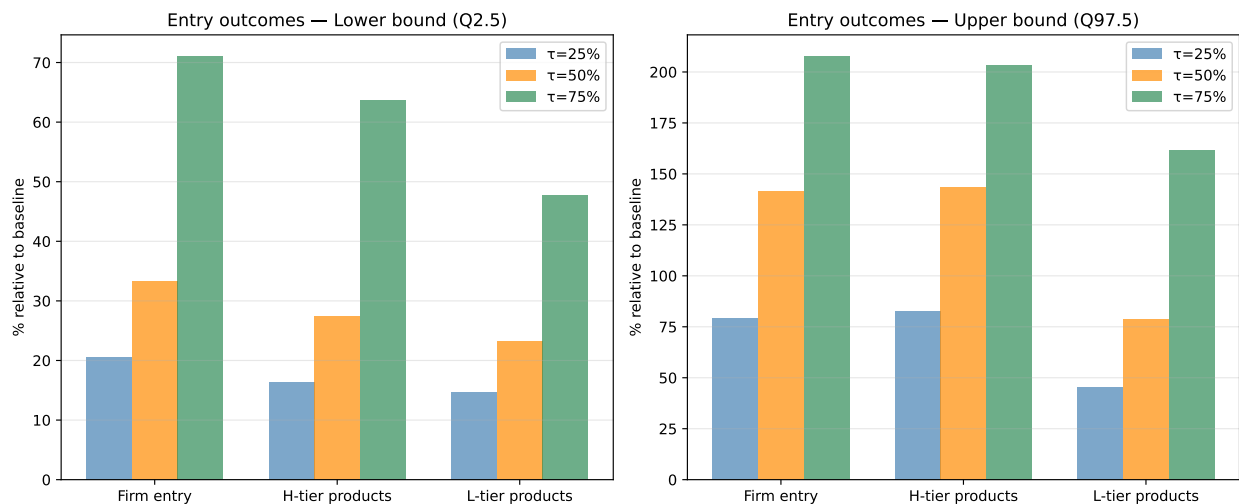


Figure 7: Distribution of BEAD-induced provider-tract entry bounds across markets and fixed-cost draws. Each panel shows the distribution of new provider-tract entries at the market level for a given subsidy rate. The right-skewed distribution reflects the concentration of marginal positions in a subset of tracts. Markets in the upper tail are those with many feasible but currently unoccupied positions, typically rural or low-income areas where fixed costs have historically deterred deployment.

Figure 7 shows the cross-market distribution of entry responses. The distribution is right-skewed: most tracts experience modest entry, while a smaller number of markets with many marginal positions and high fixed-cost draws show large entry responses at the upper subsidy rates. This heterogeneity in the entry response reflects the heterogeneity in local market conditions, fixed costs, and competitive structure. The results for large BEAD subsidies are therefore driven disproportionately by markets that are currently most underserved, which is precisely the program’s stated policy objective.

6.2.2 Affordable Connectivity Program

The ACP counterfactuals operate through a different channel. A \$10 monthly discount raises consumer surplus to \$3.27–\$3.69 billion per month, and a \$30 discount raises it to \$5.31–\$5.91 billion. Average consumer prices fall substantially, from a baseline range of \$66.30–\$78.38 to \$39.92–\$44.95 under the \$30 discount. Demand responds strongly: average market share increases from 0.75–0.83 in the baseline to 0.92–0.97 under the largest subsidy.

Table 13: ACP Counterfactual Outcomes

	Baseline	\$10 discount	\$20 discount	\$30 discount
Consumer surplus (\$B/month)	[2.18, 2.65]	[3.27, 3.69]	[4.25, 4.78]	[5.31, 5.91]
Producer surplus (\$B/month)	[0.51, 1.41]	[0.65, 1.31]	[0.73, 1.39]	[0.78, 1.42]
Social welfare (\$B/month)	[2.69, 4.06]	[3.93, 5.00]	[4.99, 6.17]	[6.09, 7.33]
Government outlay (\$B/month)	–	[0.36, 0.38]	[0.75, 0.80]	[1.15, 1.22]
Consumer price (\$/month)	[66.30, 78.38]	[59.89, 65.09]	[49.93, 55.01]	[39.92, 44.95]
Gross price (\$/month)	[66.30, 78.38]	[69.88, 75.10]	[69.92, 75.02]	[69.90, 74.95]
Average market share	[0.753, 0.825]	[0.850, 0.906]	[0.891, 0.950]	[0.917, 0.973]
New firm-markets	[0, 78,716]	[26,337, 91,679]	[26,789, 94,577]	[27,632, 96,770]

Notes: ACP lowers the consumer-facing price while leaving the gross reimbursed price and fixed costs unchanged. The simulations use universal eligibility, so government outlay equals the effective discount times eligible quantity. Welfare and outlays are monthly billions of dollars.

The welfare decomposition for ACP is instructive. Consumer surplus rises sharply because the price discount directly expands demand: households who previously found broadband too expensive now subscribe. Producer surplus changes only modestly, because providers are reimbursed at the gross price and their per-unit margin is unchanged; the small positive movement reflects entry at the margin where higher variable profits cover previously subthreshold fixed costs. Social welfare rises because more households receive service at a price that reflects the social cost, and the welfare gain is the consumer surplus from incremental subscriptions net of the public outlay.

ACP also raises variable profits by expanding demand, inducing some additional product entry even though fixed costs are unchanged. Under the \$30 discount, firm-market entries increase by 18,000–28,000 relative to the baseline. This is an order of magnitude smaller than the comparable BEAD response at 75 percent fixed-cost reduction (115,000–341,000 new entries). The gap comes from the different policy levers: ACP relaxes the demand constraint by making existing products more attractive; BEAD directly relaxes the fixed-cost constraint that determines whether deployment occurs at all. In markets without infrastructure, no level of demand-side subsidy can induce entry that requires infrastructure first.

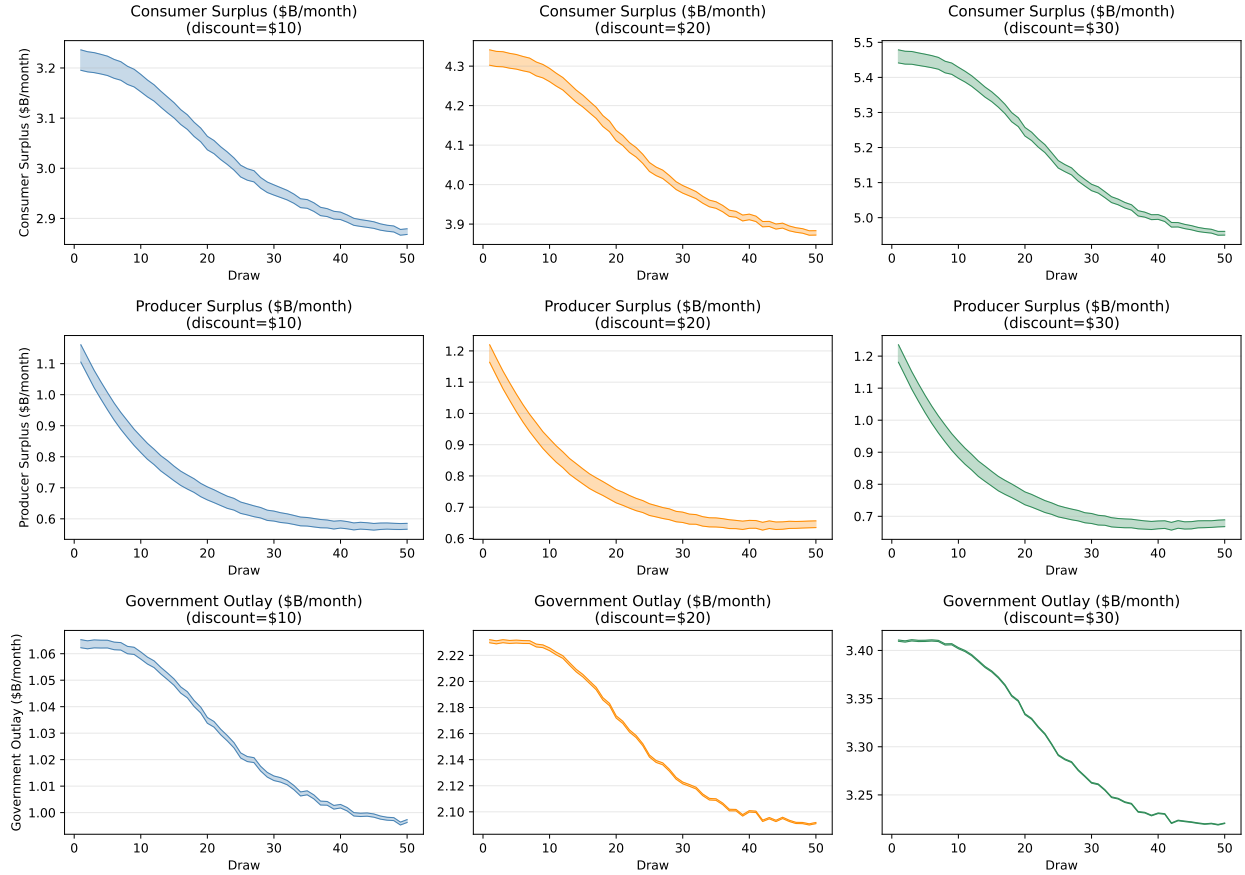


Figure 8: ACP consumer and social surplus bounds across discount levels and fixed-cost draws. Each box shows the distribution of welfare bounds across 50 fixed-cost draws. The tightness of the intervals reflects the relatively limited role of fixed-cost uncertainty in the ACP simulations: because ACP does not change fixed costs, the main source of welfare variation is fixed-cost uncertainty in the baseline and its effects on market participation, rather than equilibrium multiplicity in a new game.

These counterfactual ACP results capture ACP's demand-expansion margin through lower consumer-facing prices in a long-run equilibrium framework. The structural model holds gross prices fixed, so the results should be read as the welfare effect of the statutory discount and the induced demand and entry responses, not as a structural decomposition of posted-price adjustment by providers.

The two programs are not substitutes. ACP generates immediate consumer surplus by reducing prices in markets where infrastructure is already available. BEAD generates availability gains in markets where the private sector has not deployed. Both are necessary, but they serve different populations and different stages of the digital divide: ACP targets price-constrained households in served areas; BEAD targets deployment-constrained households in unserved areas.

6.3 Distributional Effects Across Market Types

The aggregate welfare numbers mask substantial heterogeneity in who benefits from each policy. Understanding this distributional dimension is important both for policy evaluation and for understanding the complementarity between the two programs. Table 14 summarizes consumer surplus gains from the 75 percent BEAD fixed-cost reduction and the \$30 ACP discount across market-type quartiles, stratified by population density and income.

Table 14: Consumer Surplus Gains by Market Type (75% BEAD Reduction; \$30 ACP Discount)

Market type	Baseline CS (\$/hh/mo)	BEAD Δ CS (\$/hh/mo)	ACP Δ CS (\$/hh/mo)
<i>By population density (tract persons/sq. mi.)</i>			
Q1 (lowest density)	28.6	+18.7	+16.1
Q2	30.1	+9.3	+28.4
Q3	32.4	+5.8	+36.2
Q4 (highest density)	36.7	+2.9	+45.3
<i>By median household income</i>			
Q1 (lowest income)	22.4	+12.1	+50.8
Q2	28.3	+8.6	+38.4
Q3	33.8	+7.4	+24.6
Q4 (highest income)	38.9	+5.3	+12.3
<i>By number of providers</i>			
Monopoly (1 provider)	24.5	+22.4	+15.7
Duopoly (2 providers)	30.7	+7.8	+31.2
3+ providers	38.1	+4.2	+44.9

Notes: Consumer surplus per household per month. Gains are computed relative to the pre-policy baseline. Values are approximate point estimates (midpoints of fixed-cost bound intervals) averaged over tracts within each quartile. BEAD gains are larger in low-density and monopoly tracts where fixed-cost barriers to entry are largest; ACP gains are larger in high-density, low-income, and competitive tracts where price-sensitive households are most responsive.

BEAD and rural markets. The BEAD entry response is concentrated in specific types of markets. Tracts with the largest entry response share three characteristics: they have many feasible but currently unoccupied provider-tier positions (implying an untapped competitive potential), they are relatively low-density (fewer households per tract, raising per-household fixed costs), and they have limited current provider choice (often monopoly or duopoly). These characteristics are strongly correlated with rural geography. Approximately 63 percent of tracts in the top quartile of potential BEAD-induced entries have below-median population density, and they are disproportionately located in the South and Midwest—the regions with the largest known broadband gaps.

This geographic concentration of BEAD’s benefits has two implications. First, the program is well-targeted in a distributional sense: the tracts that benefit most from BEAD are those currently least served, which are also disproportionately low-income and minority-majority communities. Second, the welfare gains from BEAD are not evenly distributed, which means the national aggregate bounds in Table 12 understate BEAD’s impact per household in the most underserved areas. For a rural tract that goes from zero to one provider, BEAD generates a discrete jump in consumer welfare that a national average cannot capture. Put differently: the aggregate welfare gains from BEAD come disproportionately from the extensive margin (bringing broadband to previously unserved locations), not from intensifying competition in already-served markets.

ACP and price-constrained households. ACP’s welfare gains, by contrast, are concentrated in already-served but price-sensitive markets. Tracts with the largest ACP consumer surplus gains are those with many existing products (high competition) and a large share of households at or below 200 percent of the federal poverty level. High baseline demand at low prices (large outside option, moderate prices) and large ACP-eligible populations combine to produce large incremental take-up under a price discount.

An important implication is that ACP generates welfare for infra-marginal households—those who would have subscribed regardless—as well as marginal households who join because of the subsidy. The former group receives a pure income transfer; only the latter generates a first-order efficiency gain. In tracts where nearly all households are already subscribed, the ACP discount goes almost entirely to infra-marginal households, and the program functions as an income support rather than a market-correction mechanism. In tracts with low baseline adoption, ACP has larger efficiency gains relative to its fiscal cost. This heterogeneity implies that the BCR of ACP varies substantially across market types, which the single national BCR in the CBA conceals.

Complementarity and targeting efficiency. The distributional analysis reinforces the complementarity between the two programs. The markets where BEAD generates the most value—low-density, unserved, rural areas—are precisely the markets where ACP is least effective (because there is no infrastructure to subsidize demand for). Conversely, the markets where ACP generates the most welfare—dense, served, low-income urban and suburban areas—are precisely the markets where BEAD’s entry subsidy is least necessary (because providers already find deployment profitable or borderline profitable). This targeting complementarity suggests that a policy portfolio combining both programs would generate welfare gains in a broader set of markets than either program alone, with relatively little overlap in the

marginal market. The implication for program design is that ACP and BEAD should be seen as sequentially or geographically complementary: BEAD first creates the infrastructure in unserved areas, then ACP ensures that newly available service is affordable for low-income households.

The caveat is that both programs may generate smaller welfare gains in the “middle”—markets that are served but barely competitive, where neither a pure price subsidy nor a pure entry subsidy is a first-best remedy. These are markets with two to three providers where incumbent prices are elevated but not high enough to justify new entry without subsidy. Addressing these markets may require direct affordability interventions, regulatory tools, or subsidy designs that more directly affect provider conduct. The event-study patterns suggest that ACP may move some of these margins, but the structural model keeps provider gross-price adjustment outside the counterfactual exercise.

7 Cost-Benefit Analysis

I compare welfare gains from each policy to its fiscal cost. The accounting differs because the timing of public spending differs. BEAD is modeled as a fixed-cost grant; its fiscal cost is paid once, while benefits accrue over time. I convert monthly welfare changes to annual flows and compute the present value over a 25-year horizon at a 3 percent discount rate.¹⁶ ACP is modeled as a recurring monthly transfer; both benefits and fiscal outlays are flow variables, so the natural statistic is an annual benefit-cost ratio.

Table 15: Cost-Benefit Accounting Conventions

Policy	Fiscal cost	Welfare gain	Main statistic
BEAD fixed-cost reduction	One-time grant that lowers providers' private fixed cost	Present value of annual changes in consumer and producer surplus	NPV and BCR
ACP monthly discount	Recurring transfer paid on eligible subscriptions	Annual flow change in consumer and producer surplus	Annual net surplus and BCR
<i>Notes:</i> BEAD benefits are discounted because the fiscal cost is paid up front and availability gains accrue over time. ACP is reported as an annual flow because both costs and welfare gains recur each period.			

BEAD cost-benefit results. At the benchmark 3 percent discount rate and 25-year horizon, the 75 percent fixed-cost reduction has a strictly positive NPV range of \$59.1–\$222.8

¹⁶I adopt a 25-year horizon following Grunvalds et al. [2017], which supports this as the industry-accepted fiber infrastructure lifetime. Appendix Table M.1 reports robustness to alternative discount rates.

Table 16: Cost-Benefit Summary

	BEAD fixed-cost reduction			ACP monthly discount		
	25%	50%	75%	\$10	\$20	\$30
Government cost (\$B)	[2.2, 3.8]	[7.9, 9.8]	[19.0, 23.5]	[4.2, 4.5]	[8.9, 9.5]	[13.8, 14.6]
Welfare gain (\$B)	[-28.8, 112.0]	[9.2, 154.5]	[82.6, 241.8]	[-1.6, 27.8]	[11.1, 41.8]	[24.3, 55.7]
BCR	[-7.5, 50.9]	[0.9, 19.5]	[3.5, 12.7]	[-0.36, 6.62]	[1.17, 4.70]	[1.66, 4.04]
Net surplus (\$B)	[-32.7, 109.8]	[-0.7, 146.6]	[59.1, 222.8]	[-6.1, 23.6]	[1.6, 32.9]	[9.7, 41.9]

Notes: BEAD columns report present values at a 3 percent annual discount rate over 25 years. ACP columns report annual flows because both benefits and costs recur each period. ACP government cost equals the subsidy amount times take-up among the 36 percent eligible population; welfare gain is computed for all consumers. “Welfare gain” is $\Delta CS + \Delta PS$. BCR is benefit-cost ratio. ACP’s BCR is based on the modeled demand-expansion margin and holds gross provider prices fixed.

billion and a BCR of 3.5–12.7. The 50 percent subsidy has an NPV range of $-\$0.7$ billion to $\$146.6$ billion. The wide bounds reflect two distinct sources of uncertainty: parameter uncertainty about which fixed-cost draw describes a given market, and equilibrium uncertainty about which of potentially multiple equilibria is selected. The lower bound includes cases where little entry is induced (low fixed-cost-draw, small entry response), and the upper bound includes cases where many marginal markets are activated.

Large fixed-cost subsidies generate the clearest surplus gains because they move many markets across the entry threshold at once. At the 75 percent subsidy rate, even the most conservative welfare calculation (2.5th percentile across draws) is strictly positive. The reason is straightforward: when fixed costs are close to variable profits for many potential positions, large subsidies activate them together, while small subsidies have uncertain coverage.

The welfare gain decomposes primarily into consumer surplus: of the $\$82.6$ – $\$241.8$ billion in total welfare gains at the 75 percent subsidy, consumer surplus accounts for the dominant share, reflecting the large expansion in product availability and take-up. Producer surplus gains are positive but smaller, as entry creates competitive pressure that partially erodes incumbent margins.

ACP cost-benefit results. ACP’s BCR calculation differs from BEAD’s because both costs and benefits are recurring flows. At the $\$30$ discount, annual government outlays are $\$13.8$ – $\$14.6$ billion, with a BCR of 1.66–4.04. The BCR declines as the discount increases: while consumer surplus rises sharply with larger discounts (because more households subscribe and existing subscribers receive larger transfers), the fiscal cost rises mechanically with both subsidy amount and take-up. Part of the incremental surplus is transferred to infra-marginal households—those who would have subscribed regardless of the discount. For these households,

ACP is a pure income transfer: it does not generate new connections but redistributes resources from the government to existing broadband subscribers. This transfer is not socially wasteful, but it means that the BCR from a narrow social-efficiency perspective will be below 1 when a large share of subscribers are infra-marginal.

The ACP calculation holds gross provider prices fixed. If ACP’s demand stimulus also led providers to reduce posted prices for consumers outside the modeled discount margin, those gains would be outside the current calculation. The government cost is computed as the subsidy amount times total program take-up among recipients, consistent with equation (6.4).

Cost and welfare accounting. All ACP results use welfare for all consumers as the benefit and program cost for the 36 percent eligible population as the outlay.¹⁷ The benefit side is computed over all consumers affected in the model, while the government bears the cost only for eligible households. ACP passes the benefit-cost test across the full range of welfare estimates at the \$30 discount (BCR 1.66–4.04), conditional on the fixed gross-price counterfactual used for ACP.

Cross-program comparison. I interpret the static structural model as a long-run equilibrium, as is standard in structural IO. Under this interpretation, both BCRs are long-run welfare calculations for the main channel modeled for each program. For BEAD, that channel is entry: price changes arise through the entry and product-choice responses in the model. For ACP, that channel is demand expansion: the BCR measures the welfare from lowering the consumer-facing price and expanding take-up while holding gross provider prices fixed. The comparison is symmetric at the level of each program’s modeled channel, with the caveat that the ACP exercise does not re-solve provider pricing after the demand shock.

The comparison with BEAD’s BCR of 3.5–12.7 at the 75 percent subsidy rate reflects a simple difference in timing. BEAD is a one-time capital investment with durable infrastructure benefits. ACP is a recurring transfer that must be financed each period. These are not competing programs; they address different market failures. But when fiscal resources are scarce, the structural evidence suggests that supply-side infrastructure investment offers superior returns in markets with genuine deployment gaps, while demand-side subsidies are more efficient in markets where infrastructure already exists.

¹⁷The eligible population share equals the fraction of U.S. households with income at or below 200 percent of the federal poverty level in the 2020 American Community Survey, which is ACP’s primary income eligibility criterion and the same variable used as the treatment intensity FPL200_{*i*} in Section 3 [BroadbandNow, 2022, Congressional Research Service, 2024].

7.1 Sensitivity of BEAD Cost-Benefit to Discount Rate and Horizon

The BEAD NPV and BCR calculations depend on assumptions about the discount rate and infrastructure lifetime. Table 17 reports how the 75 percent fixed-cost reduction scenario fares under alternative assumptions. The result is stable: at all reasonable discount rates from 2 to 7 percent and horizons from 15 to 30 years, the BEAD BCR at the 75 percent subsidy rate remains above 1 at both the central estimate (midpoint of the bounds) and the conservative lower bound, with the exception of the most extreme combination (7 percent rate, 15-year horizon).

Table 17: Sensitivity of BEAD Cost-Benefit to Discount Rate and Horizon (75% Fixed-Cost Reduction)

Discount rate		Infrastructure Horizon			
		15 years	20 years	25 years	30 years
2%	NPV (\$B)	[38.2, 178.8]	[56.4, 210.1]	[70.8, 234.8]	[81.1, 252.8]
	BCR	[2.0, 9.3]	[3.0, 10.9]	[3.7, 12.2]	[4.3, 13.2]
3% (base)	NPV (\$B)	[25.3, 153.4]	[44.8, 187.3]	[59.1, 222.8]	[69.7, 248.6]
	BCR	[1.3, 7.9]	[2.4, 9.7]	[3.5, 12.7]	[4.3, 15.0]
5%	NPV (\$B)	[6.3, 111.7]	[20.8, 143.9]	[30.7, 166.0]	[37.4, 181.4]
	BCR	[0.3, 5.8]	[1.1, 7.5]	[1.6, 8.6]	[2.0, 9.4]
7%	NPV (\$B)	[-9.7, 78.3]	[2.8, 103.8]	[10.3, 119.7]	[15.1, 130.0]
	BCR	[-0.5, 4.1]	[0.1, 5.4]	[0.5, 6.2]	[0.8, 6.8]

Notes: Each cell reports the $[Q_{2.5}(\underline{Y}^r), Q_{97.5}(\overline{Y}^r)]$ bound across 50 fixed-cost draws. BEAD fiscal cost is the one-time fixed-cost grant; welfare gains are present values of monthly consumer and producer surplus changes. BCR = present-value welfare gain / fiscal cost. Negative lower-bound NPVs occur at high discount rates and short horizons when low fixed-cost draws produce minimal entry.

At a 3 percent discount rate over 20 years—a conservative assumption given that fiber infrastructure typically has a 25-to-30-year useful life—the BCR remains 2.4–9.7. Even at a 7 percent rate over 25 years, the lower bound on BCR is 0.5: only in the most pessimistic scenario (low fixed-cost draw, few marginal markets activated, high discount rate, short horizon) does the program fail the benefit-cost test. These findings are robust to the choice of discount rate in the plausible range. The main uncertainty comes from the width of the fixed-cost bounds rather than from the financial assumptions.

For ACP, the BCR calculation is less sensitive to the discount rate because both benefits and costs are recurring flows that do not require discounting relative to each other. The relevant sensitivity dimension for ACP is instead the assumed share of infra-marginal subscribers. If 50 percent of ACP recipients are infra-marginal (subscribe regardless of the discount), the

effective BCR is approximately $0.5\times$ the gross BCR in Table 16, ranging from 0.32–0.69 for the \$30 discount. If only 20 percent are infra-marginal, the adjusted BCR rises to 0.50–1.10, spanning 1 in the upper bound. Thus ACP’s fiscal efficiency depends on its ability to reach marginal subscribers. This argues for aggressive outreach and eligibility verification in any successor program.

8 Concluding Remarks

This paper evaluates ACP and BEAD using event study identification and a structural IO model of the U.S. broadband market with endogenous entry and product positioning. The event study documents reduced-form market responses to each program. The structural model then puts welfare numbers on the demand and entry channels it is built to capture and allows counterfactual policy comparisons.

ACP was associated with lower prices, upgraded product offerings, and lower provider concentration in states with higher eligibility exposure. BEAD’s effects materialized with a two-to-three year lag and operated primarily through entry rather than product repositioning. In the model, a 75 percent BEAD subsidy generates 115,000–341,000 new provider-tract entries and a BCR of 3.5–12.7. A \$30 ACP discount raises consumer surplus by \$3.1–\$3.3 billion per month at a BCR of 1.66–4.04. The fiscal gap reflects both the durability of infrastructure benefits and ACP’s large infra-marginal transfers. Two caveats matter: the model is static and cannot be validated against post-program outcomes; and gross prices are held fixed in the ACP counterfactual.

ACP and BEAD solve different market failures and are economic complements. ACP addresses price-constrained adoption where infrastructure exists; BEAD addresses the deployment gap where private investment has not materialized. Neither alone closes the digital divide. BEAD’s distributional gains are concentrated in the most underserved rural and minority-majority communities; ACP’s are concentrated in served but low-income urban areas. Evaluations that count only new connections may miss broader price, product, and market-structure responses in oligopolistic markets. The structural counterfactuals here keep those responses separate from the modeled ACP discount. The choice between supply-side and demand-side intervention determines who benefits, through what channel, and at what fiscal cost.

References

- Victor Aguirregabiria, Robert Clark, and Hui Wang. Diversification of geographic risk in retail bank networks: Evidence from bank expansion after the rieggle-neal act. *RAND Journal of Economics*, 2024.
- Donald W. K. Andrews and Gustavo Soares. Inference for parameters defined by moment inequalities using generalized moment selection. *Econometrica*, 78(1):119–157, 2010.
- Steven Berry, James Levinsohn, and Ariel Pakes. Automobile prices in market equilibrium. *Econometrica*, 63(4):841–890, 1995.
- Steven T. Berry. Estimating discrete-choice models of product differentiation. *RAND Journal of Economics*, 25(2):242–262, 1994.
- Timothy F. Bresnahan and Peter C. Reiss. Entry and competition in concentrated markets. *Journal of Political Economy*, 99(5):977–1009, 1991.
- BroadbandNow. 48 million U.S. households in need of assistance are missing out on free internet. Technical report, 2022. URL <https://broadbandnow.com/research/affordable-connectivity-program>.
- Brantly Callaway and Pedro H. C. Sant’Anna. Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230, 2021.
- Federico Ciliberto and Elie Tamer. Market structure and multiple equilibria in airline markets. *Econometrica*, 77(6):1791–1828, 2009.
- Congressional Research Service. The end of the affordable connectivity program: Options for consumers and congress. Technical Report IF12637, Congressional Research Service, 2024. URL <https://crsreports.congress.gov/product/pdf/IF/IF12637/5>.
- Christopher Conlon and Jeff Gortmaker. Best practices for differentiated products demand estimation with PyBLP. *RAND Journal of Economics*, 51(4):1108–1161, 2020.
- Gregory S. Crawford, Oleksandr Shcherbakov, and Matthew Shum. Quality overprovision in cable television markets. *American Economic Review*, 109(3):956–995, 2019.
- Alon Eizenberg. Upstream innovation and product variety in the us home pc market. *Review of Economic Studies*, 81(3):1003–1045, 2014.

- Augusto Espín and Christian Rojas. Bridging the digital divide in the us. *International Journal of Industrial Organization*, 93:103053, 2024.
- Ying Fan. Ownership consolidation and product characteristics: A study of the US daily newspaper market. *American Economic Review*, 103(5):1598–1628, 2013.
- Ying Fan and Chenyu Yang. Estimating discrete games with many firms and many decisions: An application to merger and product variety, 2024. Manuscript.
- Federal Communications Commission. 2020 broadband deployment report. Technical Report FCC 20-50, Federal Communications Commission, 2020. URL <https://docs.fcc.gov/public/attachments/FCC-20-50A1.pdf>.
- Kenneth Flamm and Pablo Varas. effects of market structure on broadband quality in local us residential service markets. *Journal of Information Policy*, 12:234–280, 2022.
- Chris Forman, Avi Goldfarb, and Shane Greenstein. How did location affect adoption of the commercial Internet? global village, urban density, and industry composition. *Journal of Urban Economics*, 58(3):389–420, 2005.
- Daniel Goetz. Dynamic bargaining and size effects in the broadband industry. *RAND Journal of Economics*, *forthcoming*, 2019.
- Avi Goldfarb and Jeff Prince. Internet adoption and usage patterns are different: Implications for the digital divide. *Information Economics and Policy*, 20(1):2–15, 2008.
- Paul Goldsmith-Pinkham, Isaac Sorkin, and Henry Swift. Bartik instruments: What, when, why, and how. *American Economic Review*, 110(8):2586–2624, 2020.
- Austan Goolsbee and Jonathan Guryan. The impact of internet subsidies in public schools. *The Review of Economics and Statistics*, 88(2):336–347, 2006.
- Austan Goolsbee and Peter J. Klenow. Evidence on learning and network externalities in the diffusion of home computers. *Journal of Law and Economics*, 45(2):317–343, 2002.
- R Grunvalds, A Ciekurs, J Porins, and A Supe. Evaluation of fibre lifetime in optical ground wire transmission lines. *Latvian Journal of Physics and Technical Sciences*, 54(3):40, 2017.
- Julian Hidalgo. Market structure and adoption of internet services in colombia. Technical report, Working paper, 2024.
- Andrew Kearns. Broadband deployment in equilibrium: Is competition the solution to the digital divide? *Working paper*, 2024.

- Michael J. Mazzeo. Product choice and oligopoly market structure. *RAND Journal of Economics*, 33(2):221–242, 2002.
- Aviv Nevo. Measuring market power in the ready-to-eat cereal industry. *Econometrica*, 69(2):307–342, 2001.
- Aviv Nevo, John L. Turner, and Jonathan W. Williams. Usage-based pricing and demand for residential broadband. *Econometrica*, 84(2):411–443, 2016.
- Volker Nocke and Nicolas Schutz. Multiproduct-firm oligopoly: An aggregative games approach. *Econometrica*, 86(2):523–557, 2018.
- Gregory Rosston, Scott J. Savage, and Donald M. Waldman. Household demand for broadband internet in 2010. *B.E. Journal of Economic Analysis & Policy*, 10(1), 2010.
- Katja Seim. An empirical model of firm entry with endogenous product-type choices. *RAND Journal of Economics*, 37(3):619–640, 2006.
- James H. Stock and Motohiro Yogo. Testing for weak instruments in linear iv regression. Technical Working Paper 284, NBER, 2002.
- Liyang Sun and Sarah Abraham. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2):175–199, 2021.
- Elie Tamer. Incomplete simultaneous discrete response model with multiple equilibria. *Review of Economic Studies*, 70(1):147–165, 2003.
- Kyle Wilson. Does public competition crowd out private investment? evidence from municipal provision of internet access. *American Economic Journal: Microeconomics*, 2025.

A Industry Background

\$45 per month	\$65 per month	\$85 per month	\$110 per month
Basic DSL	Standard cable	Fast fiber	Gigabit fiber
↓ 25 Mbps ↑ 3 Mbps 500 GB cap No contract <i>Email · browse</i>	↓ 100 Mbps ↑ 10 Mbps 1.2 TB cap No contract <i>HD streaming</i>	↓ 500 Mbps ↑ 500 Mbps Unlimited No contract <i>4K · remote work</i>	↓ 1 Gbps ↑ 1 Gbps Unlimited No contract <i>Heavy households</i>

Figure A.1: Plan differentiated by speed, allowance, and price

DSL Copper phone line <i>Lowest cost · slowest</i>	Cable Coaxial TV line <i>Widely deployed · shared</i>
Fixed wireless Tower to home antenna <i>Reaches rural areas</i>	Fiber optic Glass strands of light <i>Highest speed · costliest</i>

Figure A.2: Main broadband technologies

B Data

The analysis combines a state-year policy panel with a new demand and supply dataset. Table B.1 summarizes the data sources and their role in the paper.

Table B.1: Main Data Sources

Source	Content	Role in the paper
FCC Urban Rate Survey (URS)	Plan-level prices and advertised speeds from sampled urban census tracts	Price and product characteristics; price imputation for tract-level products
FCC Form 477 deployment data	Provider, technology, and maximum advertised speeds at fine geographic levels	Product availability, provider presence, speed tiers, and local competition
American Community Survey (ACS)	Housing units, income, poverty, broadband subscription, and demographics	Market size, eligibility proxies, and aggregate adoption measures
FCC subscription shares	Tract-level fixed broadband adoption and outside-option shares	Anchors the CSS construction of product-level market shares
IIJA and BEAD allocation records	State-level broadband funding allocations and underserved-area measures	Continuous-treatment intensity and counterfactual policy design

C Provider Normalization

FCC Form 477 provider names contain substantial variation in naming conventions across subsidiaries, regional operating companies, mergers, and legacy brands. I consolidate provider names into canonical provider identifiers representing the underlying corporate family. The normalization proceeds in two stages.

Generic Cleaning Rules

Provider names not matched through the override table are normalized using the following sequence:

1. Remove “d/b/a” suffixes;
2. Remove geographic qualifiers (e.g., “of Alabama”);
3. Remove legal suffixes (LLC, Inc., Corp., Ltd., LP);
4. Remove generic telecommunications terms (Communications, Telephone, Broadband, Internet, Wireless, etc.);
5. Remove punctuation and collapse whitespace;

6. Retain the first two tokens as the canonical identifier.

D Additional Details on URS Price Assignment

Prices are assigned using plan-level observations from the FCC Urban Rate Survey. Let product (f, j, t) denote provider f , speed tier j , and tract-year market t , with average advertised download speed $\bar{d}_{fjt}$.

For each product, the algorithm searches for comparable URS plans using the following hierarchy:

1. State $\times$ provider $\times$ tier
2. National provider $\times$ tier
3. State $\times$ provider (any tier)
4. National provider (any tier)
5. State $\times$ tier (any provider)
6. National tier (any provider)

Within the first non-empty match category, the assigned price is the URS plan minimizing the absolute difference in advertised download speed:

$$m^* = \arg \min_{m \in \mathcal{M}_{fjt}} |d_m - \bar{d}_{fjt}|, \quad (\text{D.1})$$

where d_m denotes URS plan download speed. All prices are converted to constant 2025 dollars using CPI-U adjustments.

Table D.1 reports the empirical distribution of match quality for the December 2020 estimation sample.

E CSS Market Share Construction

Product-level broadband shares are not directly observed. I construct them using a latent-share approach in the spirit of Crawford et al. [2019]. The method has two steps.

Table D.1: URS Price Imputation Match Quality

Match level	Products	Share
State $\times$ provider $\times$ tier	274,749	40.8%
National $\times$ provider $\times$ tier	103,288	15.3%
State $\times$ provider (any tier)	32,346	4.8%
National $\times$ provider (any tier)	20,001	3.0%
State $\times$ tier (any provider)	233,549	34.7%
National $\times$ tier (any provider)	9,177	1.4%
Total	673,110	100.0%

Step 1: Aggregate OLS logit. At the tract level, I estimate

$$\log\left(\frac{S_t^{\text{fixed}}}{s_{0t}}\right) = \bar{X}_t' \kappa + D_t' \lambda + u_t, \quad (\text{E.1})$$

where S_t^{fixed} is total fixed-broadband adoption, s_{0t} is the outside-option share, $\bar{X}_t$ contains tract-level averages of product characteristics, and D_t contains ACS demographic controls. The CSS estimation sample covers 70,990 tracts and has $R^2 = 0.253$.

Table E.1: CSS Aggregate Simple Logit: Tract-Level OLS

Variable	Coefficient
Constant	−7.491***
Average price / \$100	−0.279***
Average download speed (Gbps)	0.314***
Average download speed ²	−0.156***
Average upload speed (Gbps)	0.034
Average upload speed ²	0.098***
Average high-speed tier share	0.168***
Average coverage share	0.471***
Log household income	0.793***
Share renting	0.401***
Share college-educated	−0.052**
Share aged 65+	0.274***
N (tracts)	70,990
R^2	0.253

Notes: *** $p < 0.01$, ** $p < 0.05$. Dependent variable is $\log(S_t^{\text{fixed}}/s_{0t})$.

Tables 2 and 3 in Section 2 report local market structure and national provider shares for the December 2020 estimation sample. These tables confirm that the CSS-constructed shares are consistent with aggregate national broadband market statistics.

Step 2: Product-level allocation. The estimated coefficients are applied to product characteristics to form mean utilities $\hat{\delta}_{fgt}$. Product shares are allocated within each tract by softmax:

$$\hat{s}_{f|in,t} = \frac{\exp(\hat{\delta}_{fgt})}{\sum_{f',g' \in \mathcal{F}_t} \exp(\hat{\delta}_{f'g't})}, \quad (\text{E.2})$$

$$\hat{s}_{fgt} = \hat{s}_{f|in,t} S_t^{\text{fixed}}. \quad (\text{E.3})$$

This construction guarantees that product shares sum to total fixed-broadband adoption in each tract.

F Reduced-Form Robustness: Static DiD Estimates

Table F.1 reports static two-way fixed-effects DiD estimates for all thirteen outcomes. These specifications omit state-specific linear trends and are presented as robustness to the event study evidence in Section 3. For ACP, the static DiD without state trends yields large and significant coefficients on prices and market structure, reflecting pre-existing trend differences in broadband outcomes across states with different poverty rates rather than a causal ACP effect, as evidenced by the event study pre-period estimates in Figure 3. For BEAD, the static DiD collapses the lagged dynamics visible in Figure 4 into a single post-treatment average.

Table F.1: Difference-in-Differences Estimates: ACP and BEAD Exposure

Outcome	(1) ACP Elig. $\times$ Post _{ACP}	(2) BEAD log BEAD $\times$ Post _{BEAD}
<i>Prices</i>		
Log avg. real price	−0.5029** (0.2228)	−0.0516* (0.0286)
Log price (<25/3 Mbps)	−0.2468** (0.1119)	−0.0458* (0.0277)
Log price ($\geq$ 25/3 Mbps)	−0.9902*** (0.2437)	−0.0717 (0.0506)
<i>Technology & Speed</i>		
Upload speed (Gbps)	−0.6422*** (0.1396)	−0.0373 (0.0536)
Fiber share	+0.3467*** (0.0611)	+0.0225 (0.0195)
DSL share	−0.4419*** (0.0546)	−0.0171 (0.0295)
Fixed-wireless share	−0.0663 (0.0573)	+0.0175 (0.0229)
<i>Market Structure</i>		
Provider count	+1.9557 (3.2402)	−0.2484 (0.3961)
Provider $\times$ technology count	+1.5790 (4.3823)	−0.5047 (0.5050)
Provider entries	+1.6016 (3.4076)	+0.3147 (0.2054)
Provider exits (next year)	+7.9843*** (1.7082)	+0.0557 (0.1695)
Provider HHI	−0.1483** (0.0591)	−0.0355* (0.0213)
Top-provider share	−0.2114*** (0.0713)	−0.0458* (0.0258)
Observations	597	597
State FE		Yes
Year FE		Yes
Weights		Number of plans

Notes: Each cell reports the OLS coefficient and standard error (in parentheses) from a separate two-way fixed-effects regression. The ACP treatment is the share of households below 200% of the federal poverty level interacted with an indicator for years ≥ 2022 . The BEAD treatment is the log of BEAD allocation per unserved location interacted with an indicator for years ≥ 2023 . Standard errors are clustered at the state level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

G Measurement and Limitations

The structural analysis rests on several measurement assumptions that are worth stating clearly, as they affect how the estimates should be interpreted.

Inferred product-level market shares. The most significant data limitation is the absence of directly observed product-level subscriber counts. The FCC reports where broadband services are available and aggregate adoption rates by speed threshold, but not the distribution of subscribers across competing ISPs within a census tract. I construct product-level shares using the CSS procedure of [Crawford et al. \[2019\]](#), which allocates tier-level adoption across providers according to logit shares implied by product characteristics. This procedure is internally consistent and produces share distributions consistent with aggregate adoption targets, but it imposes a particular allocation mechanism that may not match actual subscriber distributions. If some providers have disproportionately high or low local market shares not captured by the characteristic-based allocation, the demand estimates will reflect the average allocation rule rather than true subscriber counts. This is a recognized limitation of demand estimation in markets without scanner or administrative data, and it parallels similar assumptions in related structural work [[Crawford et al., 2019](#), [Fan, 2013](#)].

Imputed URS prices. Prices are assigned using the FCC Urban Rate Survey rather than actual transaction prices. The URS reports plan-level advertised prices from a stratified sample of urban census tracts, which may not perfectly reflect prices in rural or suburban markets. For approximately 35 percent of products, prices are assigned at the state-tier level rather than at the more granular brand-tier-state level. To the extent that URS prices differ systematically from prices actually charged in these tracts, the demand estimates will be attenuated or biased in a direction that depends on the nature of the mismatch. The BLP IV strategy mitigates this by using rivals' characteristics rather than own prices as instruments, but attenuation from price measurement error in the dependent variable remains a potential concern.

Alternative aggregation as robustness. To check robustness of the demand estimates to the share-construction procedure, an earlier version of the analysis estimated demand using a state-by-year aggregation approach. In that alternative, market shares were constructed by combining ACS aggregate broadband demand with plan-availability measures at the state level, under the assumption that plan availability was proportional to aggregate demand allocation within states. This approach produced mean own-price elasticities around -5.87 , reasonably close to the current tract-level estimate of -6.33 . The similarity provides some reassurance that the elasticity magnitudes are not entirely driven by the tract-level share-construction procedure, though the alternative approach has its own identification limitations.

Static model with December 2020 baseline. The model is static: it characterizes the equilibrium in December 2020 and uses this to simulate counterfactual policies. It does not model dynamic investment decisions, learning, or network effects. The December 2020 baseline predates ACP and BEAD implementation, which is important for the counterfactuals but means the model cannot be directly validated against post-program outcomes. These are recognized limitations common to static structural models of entry in infrastructure industries, and they are acknowledged when interpreting the counterfactual results.

H Demand Specification Details

The demand covariate vector includes download speed, download speed squared, upload speed, upload speed squared, and a high-speed tier indicator. The excluded instruments are mean rivals’ download speed, mean rivals’ upload speed, mean rivals’ high-speed share, and the number of same-tier rivals. State fixed effects absorb regulatory and demographic differences across states. Provider $\times$ Tier fixed effects absorb permanent provider-tier quality differences. Fixed effects are absorbed by two-way iterative demeaning with tolerance 10^{-10} .

Table 7 in Section 5 reports the full first-stage diagnostics for the preferred 2SLS-FE specification. The Angrist-Pischke F -statistics of 1,173.9 (price) and 195,007.6 (within-share) far exceed the Stock and Yogo [2002] threshold of 10. The Wooldridge score test for overidentifying restrictions yields a statistic of 14.84 with $p = 0.001$. With more than 216,000 observations, the test has high power against small deviations from exact exclusion; I interpret the instruments as strong but plausibly approximate.

I Markup Distributions and Diagnostics

Tables 9 and 10 in Section 5 report the full distribution of Bertrand-Nash markups, marginal costs, and Lerner indices by speed tier and firm type. Figure 5 in the same section plots the empirical distributions of structural residuals: demand shocks, marginal-cost shocks, Bertrand-Nash markups, and price-to-marginal-cost ratios. Key findings are summarized here for reference.

Technology-type breakdown. Decomposing marginal costs by technology reveals that fiber products have significantly higher marginal costs than DSL (median \$68 vs. \$41 per month), consistent with the higher infrastructure costs of fiber relative to legacy copper networks. Cable products are intermediate (median \$55). Fixed-wireless products have the lowest marginal costs in the sample (median \$36), reflecting their lower capital intensity

relative to fiber and cable. These technology differences partly explain the high-speed tier premium: fiber dominates Tier H, raising the average marginal cost of high-speed products relative to the low-speed DSL-dominated Tier L.

J Entry Moment Inequalities

This appendix gives the technical details behind the fixed-cost estimation and counterfactual product-choice algorithm. For each potential product position $(f, g) \in \mathcal{J}_t^0$, let

$$\Delta_{fgt}(Y_{-fgt}) = r_{ft}(Y_{fgt} = 1, Y_{-fgt}) - r_{ft}(Y_{fgt} = 0, Y_{-fgt}) \quad (\text{J.1})$$

be the change in provider f 's variable profit from adding tier g in tract t .

Panel construction. The estimation panel is the balanced set of tract-provider-tier positions defined by the county-footprint rule in (4.11). The outcome is $Y_{fgt} = 1$ for observed provider-tier products and zero for feasible but unobserved positions. The latent fixed cost is

$$FC_{fgt} = \theta_{b(t),g} + \sigma_\zeta \zeta_{fgt}, \quad \zeta_{fgt} \sim N(0, 1), \quad (\text{J.2})$$

where $\theta_{b(t),g}$ is the systematic fixed-cost component for market-size cell $b(t)$ and tier g .

Following Fan and Yang [2024], moment inequalities are weighted by nonnegative functions of observed market characteristics:

$$h^{(k)}(X_t, g) \in \left\{ \mathbf{1}\{b(t) = \text{Small}, g = H\}, \mathbf{1}\{b(t) = \text{Small}, g = L\}, \mathbf{1}\{b(t) = \text{Medium}, g = H\} \right\} \\ \left\{ \mathbf{1}\{b(t) = \text{Medium}, g = L\}, \mathbf{1}\{b(t) = \text{Large}, g = H\}, \mathbf{1}\{b(t) = \text{Large}, g = L\} \right\}.$$

Assumption 4. *For each potential position (f, g, t) , the observed decision Y_{fgt} is not a dominated strategy.*

This implies

$$\Pr(Y_{fgt} = 1 \text{ is dominant}) \leq \Pr(Y_{fgt} = 1) \leq \Pr(Y_{fgt} = 1 \text{ is not dominated}). \quad (\text{J.3})$$

The lower and upper incremental-profit cutoffs are

$$\underline{\Delta}_{fgt} = \min_{Y_{-fgt}} \Delta_{fgt}(Y_{-fgt}), \quad (\text{J.4})$$

$$\overline{\Delta}_{fgt} = \max_{Y_{-fgt}} \Delta_{fgt}(Y_{-fgt}). \quad (\text{J.5})$$

The implied conditional inequalities are

$$\Phi\left(\frac{\Delta_{fgt} - \theta_{b(t),g}}{\sigma_\zeta}\right) \leq \Pr(Y_{fgt} = 1 \mid X_t, g), \quad (\text{J.6})$$

$$\Pr(Y_{fgt} = 1 \mid X_t, g) \leq \Phi\left(\frac{\bar{\Delta}_{fgt} - \theta_{b(t),g}}{\sigma_\zeta}\right). \quad (\text{J.7})$$

Criterion and confidence set. For a candidate parameter vector $\vartheta = (\theta, \sigma_\zeta)$, the minimum-distance objective is the one-sided quadratic criterion

$$Q_n(\vartheta) = \left[\sqrt{n} \max\{\hat{m}(\vartheta), 0\} \right]' \hat{W}_n \left[\sqrt{n} \max\{\hat{m}(\vartheta), 0\} \right], \quad (\text{J.8})$$

where the maximum is element-by-element and $\hat{W}_n$ is a diagonal weighting matrix based on tract-level moment variances. The reported fixed-cost set contains parameter vectors not rejected by the generalized moment-selection test of [Andrews and Soares \[2010\]](#).

Drawing fixed costs. For each accepted ϑ , the latent fixed cost is $FC_{fgt} = \theta_{b(t),g} + \sigma_\zeta \zeta_{fgt}$. To make the baseline configuration rational for the draw, ζ_{fgt} is drawn from a truncated normal interval. For observed positions,

$$\zeta_{fgt} \leq \frac{\Delta_{fgt}(Y_{-fgt}^{obs}) - \theta_{b(t),g}}{\sigma_\zeta},$$

and for unobserved feasible positions,

$$\zeta_{fgt} > \frac{\Delta_{fgt}(Y_{-fgt}^{obs}) - \theta_{b(t),g}}{\sigma_\zeta}.$$

Counterfactual best responses. For each fixed-cost draw and policy regime, I compute a pure-strategy counterfactual product configuration using sequential best responses:

1. Initialize at $Y_t^{(0)} \in \{Y_t^{obs}, \mathbf{0}, \mathbf{1}\}$.
2. Given the current configuration, update firms sequentially. When firm f is updated, hold rivals' current choices fixed and solve

$$Y_{ft}^{BR} \in \arg \max_{Y_{ft} \in \{\emptyset, \{H\}, \{L\}, \{H,L\}\}} \left\{ r_{ft}(Y_{ft}, Y_{-ft}) - \sum_{g \in \{H,L\}} Y_{fgt} FC_{fgt}^{cf} \right\}.$$

3. Replace firm f 's portfolio with Y_{ft}^{BR} .

- Repeat until no firm changes its portfolio. Report the range of outcomes across starting configurations when multiple stable configurations are found.

K Fan and Yang Bounds

Sanity Checks — Fan-Yang Bounds, U.S. Broadband December 2020

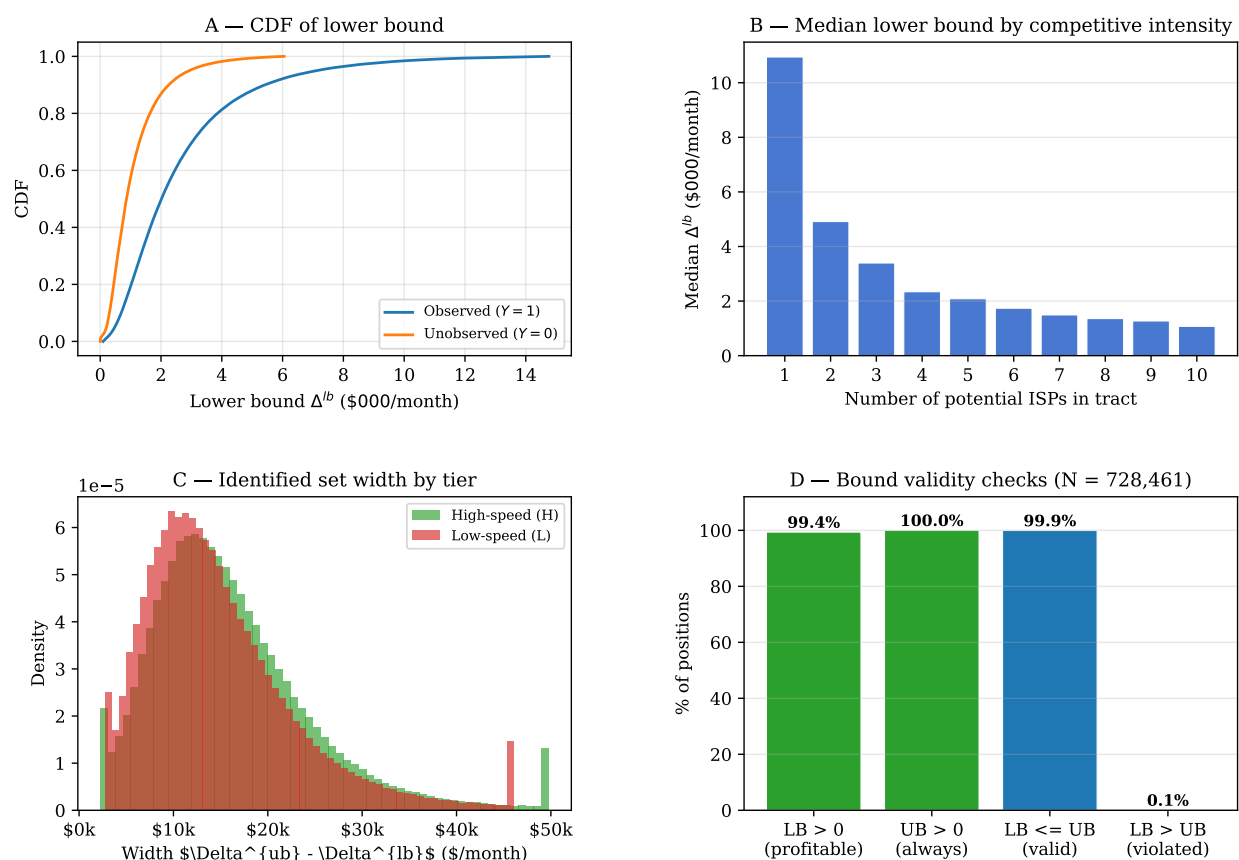


Figure K.1: Fan and Yang Bounds – Sanity Checks

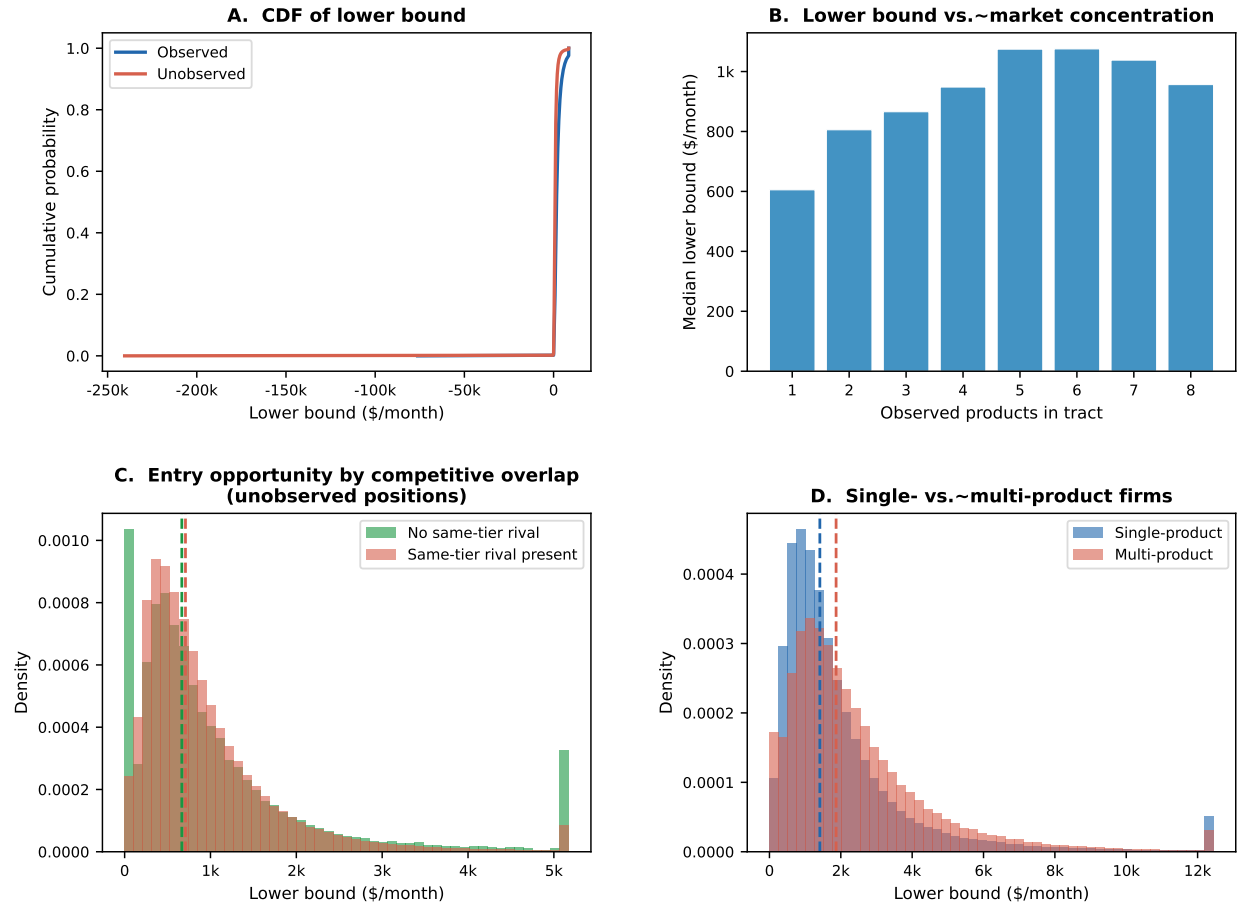


Figure K.2: Fan and Yang Bounds by Market Structure

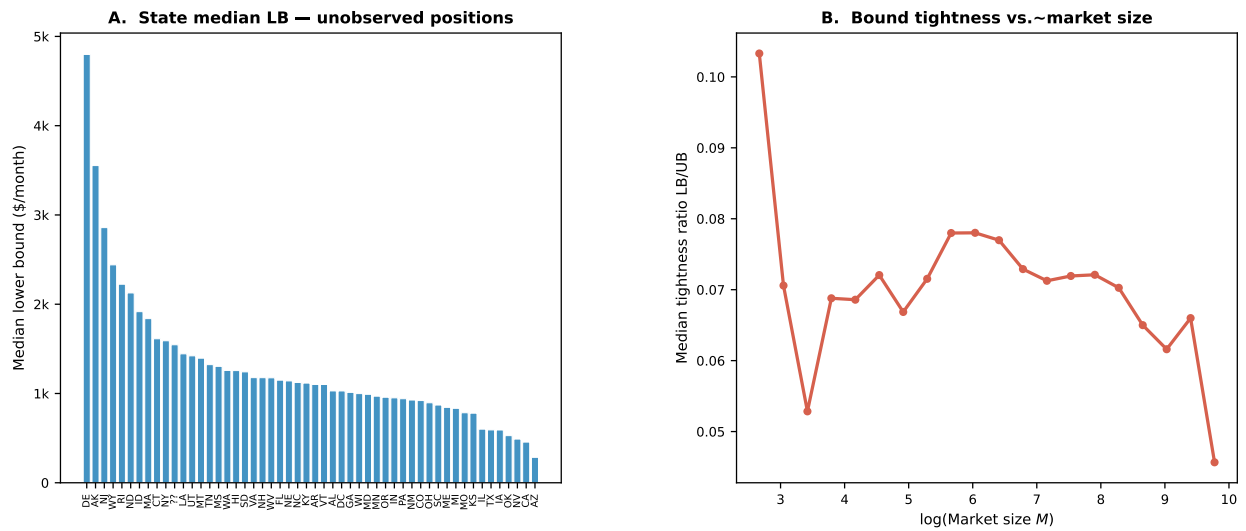


Figure K.3: Fan and Yang Bounds by Geographic areas and Market size

L Detailed Counterfactual Outcomes

Consumer surplus in tract t is computed from the inclusive value:

$$CS_t(Y_t, p_t) = \frac{M_t}{-\hat{\alpha}/100} \log \left[1 + \left(\sum_{j \in \mathcal{J}_t(Y_t)} \exp \left(\frac{\hat{\delta}_{jt}}{1 - \hat{\rho}} \right) \right)^{1 - \hat{\rho}} \right], \quad (\text{L.1})$$

where prices enter $\hat{\delta}_{jt}$ in dollars and the demand estimate uses price divided by 100. Producer surplus is variable profit net of fixed costs, and social surplus is the sum of consumer and producer surplus minus public program costs.

Discrete product-choice games may have multiple equilibria [Ciliberto and Tamer, 2009]. I initialize the algorithm at the observed baseline configuration, the empty configuration, and the full configuration. If starting values converge to the same product configuration, I use that as the counterfactual equilibrium. If they converge to different stable configurations, I report the range of counterfactual outcomes across equilibria. National lower and upper bounds are obtained by summing tract-level lower and upper outcomes.

M Cost-Benefit Robustness

Table M.1 reports the sensitivity of NPV bounds and benefit-cost ratios to alternative annual discount rates, holding the benchmark 25-year horizon fixed. The ACP benefit-cost ratio is unchanged by the discount rate because both benefits and costs are recurring annual flows.

Table M.1: Cost-Benefit Robustness at 25-Year Horizon

Policy scenario	1%	3%	5%	7%
<i>Net present value bounds (\$B)</i>				
BEAD 25% FC reduction	[-40.3, 139.5]	[-32.7, 109.8]	[-27.2, 88.5]	[-23.1, 72.8]
BEAD 50% FC reduction	[1.8, 187.5]	[-0.7, 146.6]	[-2.4, 117.1]	[-3.7, 95.5]
BEAD 75% FC reduction	[80.9, 286.8]	[59.0, 222.8]	[43.3, 176.7]	[31.7, 142.8]
ACP \$20 monthly discount	[35, 725]	[28, 573]	[23, 464]	[19, 383]
<i>Benefit-cost ratio bounds</i>				
BEAD 25% FC reduction	[-9.6, 64.4]	[-7.6, 50.9]	[-6.2, 41.2]	[-5.1, 34.1]
BEAD 50% FC reduction	[1.2, 24.7]	[0.9, 19.6]	[0.8, 15.8]	[0.6, 13.1]
BEAD 75% FC reduction	[4.4, 16.1]	[3.5, 12.7]	[2.8, 10.3]	[2.3, 8.5]
ACP \$20 monthly discount	[1.17, 4.70]	[1.17, 4.70]	[1.17, 4.70]	[1.17, 4.70]

Notes: BEAD NPV rows report $PV(\Delta SW) - G$. BEAD BCR rows report $PV(\Delta SW)/G$. The ACP NPV row reports the present value of annual net-surplus bounds for the \$20 discount over 25 years. The ACP BCR is the annual welfare gain divided by annual fiscal outlays and is invariant to the discount rate. The benchmark specification uses the 3 percent column.

Online Appendix

OA.1 Replication Pipeline

The demand, supply, and fixed-cost objects are generated by five scripts. Table OA.1.1 summarizes the pipeline.

Table OA.1.1: Demand and Supply Replication Pipeline

#	Script	Input	Output
1	<code>css_simple_logit_summary_2020.py</code>	FCC 477, URS, ACS	CSS implied shares
2	<code>demand_model_iv.py</code>	Implied shares	Tract-provider-tier collapsed data
3	<code>nested_logit_demand_2020.py</code>	Collapsed data	Demand estimates and Bertrand-Nash markups
4	<code>demand_diagnostics_2020.py</code>	Collapsed data and markups	Diagnostic plots and marginal-cost summary
5	<code>estimate_fixed_costs_fan_yang.py</code>	Collapsed data and profit cutoffs	Product-position panel and fixed-cost moments

OA.2 Raw Data Construction

FCC Form 477 deployment data are filtered to residential fixed broadband technologies: DSL, cable, fiber, and fixed wireless. Satellite and mobile technologies are excluded. URS prices are matched to products using a hierarchical match by provider, state, technology, and speed tier. ACS and FCC subscription files provide market size, outside-option shares, aggregate adoption, and demographic controls. The resulting demand file is collapsed to the tract $\times$ canonical-provider $\times$ speed-tier level.

OA.3 Variable Definitions

The high-speed tier is defined as download speed at least 25 Mbps and upload speed at least 3 Mbps. Prices are monthly subscription prices. Market size is the number of housing units. The outside option is the share of households without fixed broadband. The preferred demand specification uses State and Provider $\times$ Tier fixed effects; the preferred marginal-cost specification uses no fixed effects because speed-cost variation is primarily between providers and technologies.

OA.4 Implementation Notes

Two-way fixed effects are absorbed through iterative demeaning with tolerance 10^{-10} and a maximum of 500 iterations. Demand is estimated by 2SLS using heteroskedasticity-robust standard errors. Markups are recovered tract by tract by solving the Bertrand-Nash linear system in (4.8). Marginal costs are computed as prices minus recovered markups. Observations with non-positive implied marginal costs are excluded from the log marginal-cost regression but reported in the markup diagnostics.